

Nine Minutes Early

A Techno-Thriller

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The companies Traceframe, Alero, Vesper, Lattice, Banca Rocca, and the Piedra Sola Energy Campus; the software systems Vellum, Murmur, and Proofbox; the Cable Rojo club; and every act attributed to them are wholly fictional.

AI use: artificial-intelligence tools were used in drafting and editing this book. Every creative decision, revision and final approval is the author's own.

First edition, 2026. Development version, August 2026.

Published by Jacob Verhoeks

For those who keep the archive, and those who sign their names.

Twelve days before the largest zero-day auction of the year settles in Patagonia, a platform engineer in Buenos Aires finds four quiet pods doing work nobody will name.

Their receipts are valid. Their signatures are real.

One of them certifies a test environment nine minutes before that environment exists.

Somebody has already decided what the market will call true.

They are not spies.

They will have to do.

Prologue: Before the Test

Piedra Sola, Río Negro. Three weeks before.

The wind came across the plateau from the west at forty kilometres an hour, and Lucia Barrenechea walked the loading bay with one hand flat on the rail because the door she had propped open would not stay where it was left. Nothing in Río Negro stayed where it was left. She had told that to a courier from Buenos Aires once and he had written it down as if it were local colour.

Three cases arrived on the Thursday delivery, sealed in grey polymer with Alero's custody tags threaded through the hasps. Witness appliances. Lucia signed for them at 16:40 under the floodlight above the bay, because the light was for exactly this: proof of condition at handover. The driver held out a tablet for her countersignature and kept his other hand in his pocket against the cold.

"Battery conditioner in number two ran during transit," he said. "It's noted on the manifest. They run warm. Nothing to check."

Lucia read the manifest. There was no note about a battery conditioner.

She said so.

The driver looked at the screen, then at the case, then at the

wind moving grit across the apron, and chose the road. "It's noted somewhere," he said. "They're sealed, señora. Your job starts when you break them."

Her job had started eleven years earlier, when she had come back from Neuquén with a thermal engineering degree and a rule she had never needed to write down: equipment tells you its history if you let it sit still long enough. She put the three cases on the inspection bench in the annex, spaced apart, and let them sit still.

At 17:20 she laid the back of her fingers against case two.

Warm.

Not sun-warm and not engine-warm. The cases had ridden in the unheated section of the truck for six hours from Bariloche, and the annex sat at fourteen degrees, and case two returned heat like a laptop that had been working in a bag. Cases one and three were at annex temperature. Case two was at thirty-one degrees and climbing, very slowly, the way a machine gives up its stored work.

She photographed the manifest. She photographed the thermometer. She wrote the time on the whiteboard with the grease pencil that lived on a string, because the annex lost network whenever the wind decided the dish had heard enough.

The trial was scheduled for that night. Alero's coordination message gave her a window between 22:00 and 23:00, a target environment to power, and a list of observations so short it was almost an insult. Power the twin. Confirm telemetry. Touch nothing else. The appliance crews would do their own work behind their own seals, in the hall she was contracted to maintain but never enter.

Lucia had spent a decade around energy customers who

modelled reservoirs and grid demand and the fatigue life of turbine blades. Confidentiality was normal in that business. Everyone's data was someone else's headline. But those customers sent her failure modes and asked questions. Alero sent her a window and a signature line.

At 22:14 she brought the target environment up.

It was a digital twin, though nobody used the word honestly. To Lucia it was a rack of instrumented compute pretending, with great precision, to be something else: a fictional product estate with services that started in a scripted order and sensors that reported loads no physical machine would ever feel. She watched the boot sequence settle, confirmed telemetry flow to the collection bus, and initialised the run clock. 22:14:02. She wrote it on the whiteboard because the whiteboard did not need the network.

At 22:31 the receipt printer in the annex chattered.

Nobody had told her there would be a printer. It was an ugly office device on a shelf above the patch panels, plugged into a circuit she had not energised, and it produced three slips while she stood there with a screwdriver in her hand. Quorum certified, the slips said. Effect confirmed. Custody continuous. Each carried signatures she did not recognise and a timestamp block that began:

22:05:47.

Nine minutes before she had powered anything.

She checked the run clock first, because you always check your own instrument before you accuse the world. The run clock agreed with the whiteboard and with her watch. Then she checked the printer's own menu, thinking some regional setting had shifted its stamps into another zone. The printer thought it was exactly where it was.

The customer line answered on the second ring, which told her somebody had been waiting on it. A man with a Madrid accent and the patience of someone reading from a card called it clock skew. He said distributed systems disagreed about time all the time. He thanked her for flagging it. He asked her, in the same breath, to countersign the physical delivery record for the three appliances, confirming receipt in good order and condition, so that custody could be closed out cleanly before the trial report issued.

"You received them sealed," he said. "That is all the countersignature covers."

"That is not all it covers," Lucia said. "It covers the condition I received them in. One of them was warm."

The pause had a shape she recognised from warranty arguments: a man deciding how much of her he needed.

"The seal is intact," he said finally. "Internal state is outside facility scope. Please countersign, señora. The trial closes tonight."

She countersigned one line and not the other. In the condition field she wrote: *Seals intact. Case 2 warm on arrival at 16:40, ambient 14 °C. Manifest makes no note of conditioner.* She wrote the time she had broken each seal, and only that. Let the document say what she knew and nothing she suspected. A record that claimed too much was worth exactly what its weakest claim was worth, and she had watched lawyers discover that arithmetic on other people's behalf.

Afterwards she swept the annex. The packing crates went into the compactor behind the maintenance shed, standard disposal, except that Alero's disposal bins sat outside her custody contract, which meant anything she threw away stopped being hers and started being evidence she had surrendered. So

she cut the thermal strips from the inside of the crate lids first. Battery-conditioner or not, a strip is cheap chemistry and honest: it remembers heat the way wax remembers a thumbprint, in bands, in order, in colours it cannot be talked out of.

Case two's strip showed a history the manifest did not mention. Powered, cooled, powered again. Not one transit event. Several.

She folded the strips into the zip pocket of her jacket, meaning to log them properly in the morning when the network came back, and went home along the plant road with the wind shouldering the car towards the verge. Below the road the valley lights of the town sat in their usual handfuls. People lived here. Trucks came through at all hours with loads that had paperwork. In eleven years she had learned that the interesting question was almost never whether machines behaved. Machines behaved. The interesting question was who had already been paid to expect it.

In the annex, case two cooled through the small hours, giving up its heat by degrees into air that smelled of dust and cold solder.

In Lucia's jacket, on the hook by her door, the strip in the zip pocket was not cooling at all. At two in the morning it changed colour again, one band over, recording a fresh band of warmth from an appliance that was, by every document in the world, sealed and asleep.

She did not see it happen.

She found it in the morning, held it up to the kitchen window, and understood that she was now holding the only honest witness anyone had sent her.

Part One: Human in the Loop

Buenos Aires, twelve days before settlement.

Chapter 1: Nine Minutes Early

The receipt was committed at 04:13:08 on a Thursday morning in August, for a test environment that came into existence at 04:22:41.

Sofía Alvarez learned this in the other order, because that is how incidents arrive. First the pager, at 04:23. Then the dashboard. Then, much later, the order of events itself, which turned out to be the entire problem.

The alert was a burn-rate page from Traceframe's regional cluster, the kind she usually cleared with half her attention before the kettle boiled. Service level objectives did not respect Buenos Aires nights. But the summary line was wrong in a way she had not seen before, and she had read eleven years of these lines.

Confidential workload: verification suite completed. 100% pass rate. Duration 0m 41s.

Completed. Not failed, not degraded, not flapping between green and amber the way test suites did when they were honest. Completed, with a pass rate of exactly one hundred per cent, against a customer whose workloads never reported anything at all. That was the arrangement.

Traceframe sold observability to a legitimate energy-research account, the account ran confidential verification jobs on isolated capacity, and the telemetry arrived wrapped in labels she was contracted not to open. Sofía kept the pipes healthy and the contents anonymous. In four years the workloads had produced metrics like a heartbeat produces sound: proof of life, nothing else.

Outside, Palermo was at the hour when the street sweepers finished and the first delivery motorbikes had not started. Her building's WhatsApp group was quiet. The window over her desk held the wet green smell of the plaza after rain, and the humidity sat on the laptop's aluminium like a second casing.

She opened the workload's namespace.

Four pods, running since 03:58. Low load was an understatement. Each pod idled below two per cent of its CPU request, memory flat, network negligible. They were not crunching anything. They were arranged across three worker nodes the way witnesses arrange themselves in a courtroom: separated on purpose, close enough to see the same thing.

And they were emitting. Small payloads, every few seconds, to an endpoint outside Traceframe's estate. Compact signed statements, a few hundred bytes each. She copied one out and read it.

It was a receipt. Effect confirmed. Environment matched commitment. Cleanup verified. Custody continuous. Three signature blocks, chained. The wording was dense and legal and strangely familiar, the way a voice on the phone is familiar before you place it. She put that aside. First the arithmetic.

The receipt certified a completed verification run against a named environment: `tw-audit-07c`, provisioned under the confidential account's project. Sofía pulled the environ-

ment's own record from the control plane. Provisioning for `tw-audit-07c` began at 04:19. The container image finished pulling at 04:21. The environment registered ready at 04:22:41.

The receipt committing effect confirmed was signed at 04:13:08.

Nine minutes and thirty-three seconds before the thing it described existed.

She did what the training in her head told her to do, which was to assume she was the malfunction. Clock skew first. Distributed systems disagreed about time constantly; every engineer eventually learned that *now* was a local opinion. If the pod's clock ran nine minutes fast, or the control plane's ran nine minutes slow, the impossibility folded into an explanation.

She pulled the time-sync telemetry for all three worker nodes. Each node held lock on its upstream sources, drift under four milliseconds, no step changes in forty days. She checked the registry timestamps against the image digest log, against the load balancer's access log, against the storage layer's own write path. Four independent systems, four sets of clocks, all agreeing within milliseconds.

Retries next. A stale cache could serve an old object under a new name. She forced a cache bypass and re-read the environment record straight from source. Created 04:22:41. Immutable audit log, append-only, hash-chained: same answer.

By five she had exhausted the boring explanations, made coffee she did not drink, and opened an incident, severity three, then raised it to two when she wrote the summary sentence and read it back.

Witness receipts for confidential verification workload predate the

existence of the environments they certify.

There was one more check, and it was the one that mattered, because timestamps could be argued with but thermodynamics could not. The worker nodes exported hardware counters. CPU package power, fan duty cycles, and, on this particular hardware, an inlet temperature sensor with more honesty than most employees. She graphed the three nodes across the window from 03:00 onward.

At 03:58, when the pods started, the nodes did nothing interesting. Two per cent CPU does not warm a machine room. But the pods' own thermal signatures, inferred from the package sensors, showed brief working spikes just after start-up, then near silence. Whatever those four processes were, they had done a little real computation and then settled into signing things other people computed.

One more graph. The environment `tw-audit-07c`, born at 04:22:41, lit up the node it landed on like a match head: provisioning pull, boot storm, the short violent forty-one seconds of a test suite executing, then teardown at 04:24.

The pods had spiked at 04:05. The environment burned at 04:23.

The order was not a clock artefact. The order was physical.

Sofia sat back and let the apartment make its small sounds: the fridge, a motorbike two streets over, the building's water pump cycling somewhere below the floor. She thought about what she had, in the language she trusted. Four records claimed an effect. One record proved a cause. The cause postdated the claims. Either somebody's infrastructure was lying in a way that survived four clock domains and a thermocouple, or somebody had decided what the verification would conclude and arranged the paperwork backwards from there.

Not prediction. Prediction would have been frightening enough, some model inferring outcomes before the test ran. This was worse and also simpler. Nobody had predicted anything. Somebody had known.

She wrote the incident timeline, each entry with its evidence link, each claim sized to what its source could support. It was a good document. She felt the small professional satisfaction that always came, even at six in the morning, of a timeline that would survive hostile reading. Then she considered the escalation field, and the confidentiality wrapper around the account, and the fact that whatever she had found lived inside pipes she was paid to keep boring and contents she was paid not to read.

She called Martin Vale anyway. Toronto was six hours behind, but Martin kept incident hours the way other men kept cigarettes, and on the third ring he answered with the flat clarity of someone who had never once been asleep on call.

"Tell me you're not paging me about the region failover."

"I need your eyes and I need you to tell me I'm stupid," Sofía said. "A confidential workload just filed passing grade on a test environment nine minutes before the environment existed. The clocks are clean. The hardware counters agree with the timestamps. Martin, the receipts are older than the machine they describe."

Silence on the line. Not the silence of someone half asleep. The silence of someone reordering their morning.

"Send me the incident," Martin said. "Nothing else. And Sofía. Which account?"

"You know I can't say that on a call."

"Which account," he said again, gently, and something in his gentleness frightened her more than the graphs had.

She gave him the incident number and nothing else. After he hung up she went back to the dashboard, because the last thing an SRE does before waking the management chain is make sure the fire is where she thinks it is.

The four pods were still running. Still idle. Still emitting their small signed certainties into the dark outside Traceframe's estate, patient as jurors.

She looked at their output again, the payload she had copied out, and this time she did not put the familiarity aside. Effect confirmed. Environment matched commitment. Cleanup verified. Custody continuous. She knew this cadence. Tight clauses, narrow verbs, every sentence built to be defensible in a room where someone wanted to sue. She had written prose like this. Everyone in reliability had, at some point, written a clause designed to survive a lawyer.

But this was not her prose. And it was not a test report, either. Now that she read it as what it was, the payloads stopped looking like telemetry at all.

The pods were barely computing. Their product was not analysis.

Their product was agreement.

Chapter 2: Clock Skew

Martin called back at 12:40, which was 06:40 in Toronto and proof that he had not gone back to bed either.

“I read the timeline twice,” he said. “It’s clean. So before anyone makes this a story about ghosts, let’s kill it properly. You checked the node clocks.”

“Locked to upstream, drift under four milliseconds, no steps in forty days.”

“You checked the control plane.”

“Registry, digest log, load balancer, storage write path. Four domains, milliseconds apart. The environment was born at 04:22:41 and the receipt celebrated its first birthday at 04:13:08.”

“Then we check the clocks you don’t own,” Martin said.

That was the difference between Martin and most security engineers Sofía had worked with. Most people, given an impossibility, reached for a bigger hammer. Martin reached for a second opinion from a machine that had no reason to lie. He had built half of Traceframe’s independent audit plumbing years ago, back when he still believed infrastructure projects ended, and the part that mattered now was the regional audit cache: a write-mostly sink in São Paulo that ingested event streams over one-way feeds and stamped them with its own

time, from its own satellites, on hardware nobody in Buenos Aires could touch.

“If your nodes conspired,” he said, “the cache is where they get caught. Stream me the window. Raw events, no summaries.”

She queued the export while she ate lunch standing at the counter, which was how she ate lunch during incidents: three cold cuts of something her neighbour had left at her door with a note about her mother’s birthday, wrapped in film, still faintly warm from the pan. The building knew her schedule better than her family did.

The audit cache answered within minutes, and it convicted everyone equally. Its own stamps put the pods’ start spikes at 04:05, the receipt commits at 04:13, and the environment’s short life at 04:22:41 through 04:24. The same order. Milliseconds of disagreement between clocks, nine minutes of agreement about the crime.

“Billing next,” Martin said.

Cloud billing was the third witness, and the least romantic. Every container second on the confidential account generated metering records for a provider whose invoices survived audits and lawsuits and had never once cared about Traceframe’s internal opinions. Metering ran on its own collectors with their own time discipline, because money does not tolerate ambiguity.

She pulled the interval records for the three worker nodes and the isolated capacity pool. The billing stream showed compute consumption beginning at 03:58 for the pods, a distinct and much larger burst for `tw-audit-07c` beginning at 04:22, and nothing at all in between that could account for a completed test suite at 04:13.

“So,” she said. “Not skew.”

“Not skew.” Papers moved on his end. She pictured his kitchen table, the one with the maintainer stickers half peeled off it, his laptop angled away from the window because he complained about glare in every season. “One more. Node monotonic counters. Not wall time. The raw counter each kernel keeps since boot. If someone rewrote timestamps, the counters still count.”

“I have them. The spikes hold the order. I sent you the graphs.”

“I know. I wanted to say it out loud once, in sequence, so we both heard it.” He said it anyway, slowly, like a man reading charges. “The pods computed first. The receipts committed second. The environment existed third. The test ran fourth. Every honest clock agrees on the order and every honest clock says the order is impossible.”

Sofía dried her hands. Outside, a colectivo sighed at the stop below her window, and the smell of wet plane trees came up with the afternoon heat.

“There’s a version of this where I’m being played,” she said. “Somebody building a story for me out of real logs.”

“I thought about that on the walk.” A chair creaked. “If it’s theatre, it’s theatre with a budget. Four clock domains, hardware counters, a billing stream in another country, and an append-only audit sink I personally architected to be annoying to forge. You don’t build all that to gaslight one SRE. You build all that when the audience is lawyers.”

“Lawyers,” she repeated.

“Or auditors. Or buyers.” He stopped there, and the silence had a door in it that he did not open.

“Martin. You’ve seen this format before. I can hear it every

time you say the word receipts. You say it like it's a brand name."

This time the pause went on long enough that she checked the call had not dropped.

"In 2023 I did some contract work," he said. "Verification side. Sealed jobs, independent witnesses, signed statements about whether a thing actually happened. It was good work. It paid in the currency of feeling useful." Another pause. "The statement format I signed had the same clause skeleton as your mystery payloads. Effect confirmed. Environment matched commitment. Cleanup verified. Custody continuous. Same cadence. Same narrow verbs. Somebody took the language and grew a market on top of it."

"Whose language?"

"Mine, partly. Ours, if we're being careful." She heard him breathe. "I can't explain what I know from one contract in a way that keeps you safe, and I can't explain it in a way that keeps me employed, until I check some things that take a day or two. So here's what I need."

"Say it."

"Keep my name out of the incident. Your timeline stands on its own evidence; it doesn't need a contributor list. As far as your management chain knows, you found this alone at four in the morning because you're the kind of person who reads burn alerts properly. Which is true."

"You want to be un-involved from an incident you're actively investigating."

"I want my daughter to keep going to a school where nobody has heard my name," Martin said, without drama, the way he would report a failed disk, and Sofía understood that this sentence had cost him more than the whole morning of

clocks.

“Two days,” she said. “Then I escalate with or without you.”

“Two days.” Relief changed his voice by perhaps five per cent. He was reliable about handovers; he would not let those two days lapse. “One more thing. The pods. Don’t touch them. Don’t restart them, don’t cordon the nodes, don’t give your on-call any excuse to drain them. Right now they’re witnesses pretending to be furniture. If you disturb them, the story becomes that you broke them.”

“And if I leave them running?”

“Then whatever they certify next, they certify into a record you already flagged. That’s not nothing. That’s the only position with any leverage in it.” He paused again, and when he continued his voice had the careful flatness of a man choosing each word on the way past. “Whatever this is, two different parties are being described to each other by machines they trust. Do not let either side own the time.”

The call ended. Sofia washed her cup, dried it, and set it upside down on the rack, because procedure was what her hands did while the rest of her thought.

On the dashboard, the four pods idled below two per cent, patient as furniture. At 15:07 local time, one of them emitted a new payload. She opened it.

Effect confirmed. Environment matched commitment. Cleanup verified. Custody continuous.

A new receipt, for a new environment, provisioned under the same confidential project. She pulled the environment’s record with the cache bypassed before the page even finished loading.

Created 15:19.

Nine minutes later than the signature again. Whatever the machinery was, it was not finished. It had simply been waiting for business hours.

Chapter 3: Customer Exception

The management chain woke up at 15:30, and by 17:00 the incident had grown a video call with six people on it, five time zones and one man who mattered.

Daniel Cho joined last, from Seattle, camera on, grey light behind him that meant rain or dawn. Director of security, sixteen years at three companies, the kind of executive who read the timeline before speaking. Sofía had met him twice in person and both times had watched him ask the one question nobody else thought to ask. She had been on his side of exactly none of this before.

“Walk me through what you can prove,” he said. “Not what you suspect.”

She walked him through it. Four clock domains in agreement. Hardware counters in agreement. An independent audit sink in São Paulo in agreement. A billing stream in a third jurisdiction in agreement. Receipts older than their referenced environments, twice, nine minutes each time, like a signature notarizing a document before the ink existed.

Daniel let her finish. Then he was quiet for four full seconds, which in executive time was a speech.

“Thank you,” he said. “Nothing you’ve described is an outage, a breach, or a data exposure. I want to confirm the boundary of what happens next.”

Her manager started to outline remediation options. Suspend the workload. Cordon the nodes. Revoke the service account issuing the receipts. Reasonable verbs, all of them, the standard toolkit for making a strange thing stop.

“None of those,” Daniel said. “Sofía, are your pods authorized under a customer exception?”

“I don’t know what they’re authorized under. That’s rather my point.”

“They are.” He said it without looking away from the camera, and she watched him spend something to say it. “There is a confidential customer exception on this account covering defensive verification workloads. It predates my stewardship of it. It has been reviewed by legal, it has a named sponsor, and it authorizes exactly the class of activity you’re describing. Signed attestations about sealed test environments. I can’t tell you the customer’s name. I can’t describe what their tests verify. I can tell you that when we approved the exception, the stated purpose was defensive research verification, and nothing in your evidence contradicts that purpose.”

“Everything in my evidence contradicts the physics of that purpose,” Sofía said. “A defensive verification suite certified a result before the thing it verifies was created. Whatever the exception authorizes, it is being executed backwards.”

“Or forwards,” someone from her manager’s team said, too quietly, and then apologized.

Daniel ignored it, which was its own kind of confirmation. “Here is what the contract requires of us, and I’m going to say it plainly so we all carry it together. We provide observability.

We do not interpret customer telemetry on the customer's behalf, we do not suspend customer workloads absent a security event within our estate, and this is not one. No data left our custody. No credential in your graphs belongs to us. The signatures on those receipts are real, they belong to keys we don't hold, and the fact that I can't explain them is not, in law or in contract, ours to fix."

"So we do nothing."

"We do our jobs," Daniel said. "You restore the monitoring gaps your alerting changes created yesterday" (there had been none, and he knew it, and the sentence was a formality shaped like an instruction) "and the incident closes as anomalous telemetry, monitored, no action. The exception holder has been notified of your findings through the contractual channel. If they want us to investigate further, they will say so through the same channel."

"And if they don't?"

"Then we remain a company that keeps its contracts," Daniel said, "which is the only kind of company anyone pays to watch anything."

Nobody spoke for a moment. On the call, her manager unmuted to confirm the closing actions, and Sofia watched the incident acquire its ending the way a river acquires a border: by agreement, downstream of any rocks.

"One question," she said. "Off the record, in the sense that I'm asking it anyway."

Daniel waited.

"The exception has a named sponsor. Is the sponsor aware that the workload's output precedes its input?"

"The sponsor is aware of everything in your report," Daniel said. "It went through the channel an hour before this call.

Which is why I can also tell you this, and then I'm done talking around my own contract." He leaned towards the camera. "Nothing in your environment is going to disappear. No logs will roll over early, no dashboards will lose history, and nobody at this company will rewrite your timeline. You wrote it well. It stays exactly as you wrote it. What happens to the information inside it is not between us and the customer anymore. It's between the customer and whoever they answer to. Restore your dashboards, close the incident, and take the win you actually got, which is that you proved something impossible and it survived contact with management."

The call ended. Her manager sent a closing summary with the word *anomalous* in bold. The incident turned grey in the queue, resolved, monitored, no action, and began its journey into the archive where companies keep the things they handled correctly.

Sofía sat in the blue hour of the apartment. Somewhere below, the parrilla smoke of the block's first dinners came up the light well with the sound of a radio tuned to a news station. She had done everything right. Escalation, evidence, chain of custody, honest timelines, a director who broke no rules and hid nothing and told her the truth in the shape of a contract.

That was what frightened her. She had spent the day assuming the obstruction would arrive wearing a mask: deleted logs, a forged message, a manager with a new priority. Instead the machinery had absorbed her finding without friction, the way a payments network absorbs a disputed charge, every step compliant, every signature real, and produced at the end a world in which four pods could certify the future and the record would show that a competent professional had been heard, taken seriously and thanked.

She checked the dashboard once more before closing the lid.

The pods were still there. Low load, low light, waiting without effort. At 19:12 a third receipt crossed the wire, effect confirmed, custody continuous, for an environment whose provisioning record would begin, she had already confirmed, at 19:21.

Restore service, the contract said, to a system whose output she could prove preceded its input.

She closed the laptop. In the dark screen her own reflection held for a moment, faint and doubled, a woman signing off on a shift that had not ended at all.

Chapter 4: Signed by a Person

The invitation arrived on Friday morning through the normal channel, which was her own inbox: Traceframe's community relations group forwarding an event calendar, one line among forty. *Argentina–Israel AI Innovation Forum, Recoleta, Friday 13:00. Open to industry partners.*

She would not normally have gone. But the incident was closed, the pods were still certifying the future at a rate of roughly one receipt per business morning, and Martin had gone quiet in a way that had a shape to it. Two days, he had said. She had thirty-six hours left of them, and nothing to do with the waiting except stand in rooms where information might be walking around.

The forum filled a Recoleta hotel ballroom with the international dialect of venture capital: bad coffee, excellent air conditioning, name tags with flags. Panels on sovereign compute. A minister's video message. Clusters of people comparing time zones between Buenos Aires, Tel Aviv and Miami. Legitimate money, mostly, doing legitimate things, and Sofía wore her Traceframe badge and let the room assume she was procurement.

Noa Ben-Ami found her at the second coffee break.

“Platform reliability,” Noa said, reading her badge with an expression that suggested the badge was a sentence she was translating. “Then you are the person who keeps the pipes boring.”

“That’s the job description.”

“Noa Ben-Ami.” A card appeared between two fingers, matte cream, no title. “Market structure consultant. I work between investors here and research groups elsewhere. Sit with me? The panels are better from a distance.”

They took the small table by the window, away from the buffet. Noa ordered for both of them without consulting the menu, in Spanish that carried no apology, and waited until the waiter left before she opened the conversation proper.

“An anomalous telemetry incident closed yesterday,” she said. “Severity two, raised to two, closed as monitored, no action. Four workloads, three nodes, one very good timeline written by a senior engineer who does not appear to sleep.” She turned the coffee cup a quarter turn on its saucer. “I would like to talk about what it proves, because I think you already know what it proves and are pretending, professionally, that you do not.”

“You have access to Traceframe’s incident queue.”

“I have access to a summary. The queue itself I am politely not asking about.” Noa smiled briefly. “Here is my position, and it is not a recruitment pitch, because you are not the person I would recruit. It is context, offered to someone who found herself standing inside a market she has not been told exists.”

For the next twenty minutes, over food neither of them touched much, Noa laid out Vesper’s case the way a lawyer

lays out facts she intends to repeat under oath. Zero-days were dangerous claims. A claim that could not be tested without being received was an instrument that travelled wrapped. Generated code had uncertain training lineage; nobody could say what a model had seen, whose work it had digested, or why a particular candidate had surfaced on a particular night. Volume was its own attack: a thousand plausible reports could bury the one real finding a volunteer maintainer had time to read.

"So a market exists for these claims whether decent people participate or not," Noa said. "It settles in the dark or it settles somewhere with lights. We are the somewhere with lights. Every claim we sell carries a named human who understands it, can reproduce it and can be held responsible for it. Not a logo. A person, with a name, who signs."

"And if the model found it first?"

"Then a human finds it again, understands it and signs. Origin is a question about labour. Custody is a question about safety. We sell custody."

Sofía turned her cup to match Noa's quarter turn. "You described my incident in detail for someone outside the company."

"I described a summary of your incident."

"You also know where my salary comes from when it leaves the country."

The room noise went on around them, the panels, the laughter by the buffet, and Noa did not blink or glance away, which was its own kind of answer.

"Banca Rocca," Sofía said, very quietly.

"A respectable institution with one inconvenient client." Noa folded her hands. "Here is the offer, and I will make

it once. Vesper retains external counsel in Buenos Aires, excellent ones, who regularize foreign income and historical non-declaration every day, quietly, correctly, permanently. You would pay nothing. In exchange, when the moment comes, and it will come, you provide us the independent timing record you already hold. The regional cache. Nothing else. Not exploit contents, which you have never had. Timing."

"And if I refuse?"

"Then you remain exactly what you are now. An engineer with an anomaly she cannot report and a bank account that could end her career twice." Noa stood, collected both cups and set them on the tray, an oddly domestic gesture. "But you should know that I mean the other part sincerely. Someone must be able to sign for dangerous code. If you ever find a different answer to that problem, a better one, I will listen to it. That is not flattery. It is the only reason I am still in this business."

She left a banknote under the saucer for the waiter and walked back into the ballroom crowd, and Sofía sat alone at the window table doing honest accounting.

Because the case was good. That was the trap inside the threat. Every clause of it matched the thing she believed: that responsibility needed a name attached, that evidence needed custody, that somebody had to be able to stand up in a room and say *I checked*. Noa had taken Sofía's own private grammar and handed it back to her with a price list.

The waiter came to clear the table and paused, apologetic, holding the card terminal.

"The lady paid cash," Sofía said.

He turned the terminal towards her, mildly confused. On its screen was an authorization request, pending, for the exact

amount of lunch. The card it named was hers.

Her corporate card, which was upstairs in her desk drawer in Microcentro, where she had left it at nine in the morning.

Chapter 5: Proof of Lunch

The terminal was a mistake, or a message, and Sofía spent the taxi home deciding which.

A pending authorization could be a cloned card, and cloned cards were weather in Buenos Aires; everyone she knew had a story about a Friday night in Palermo and a Monday morning full of Neuquén. She did the boring things first, from the back seat, on her phone. Freeze the corporate card. Freeze the personal one. The bank's app made her authenticate twice and then confirmed both cards cold at 16:51.

Then she opened the expense system, because she wanted to see the charge in its natural habitat before reporting it, and the habitat was wrong.

The authorization had posted at 13:44, during lunch, at the hotel where she had sat with Noa. Amount matching the set menu. But under it, already attached to the same merchant record, sat two more entries that no clone could explain, because they were not charges. They were corroborations. A building access record: her badge, the hotel's conference wing, 12:58. A device-risk signal from Traceframe's own security stack: her phone observed on the venue network from 12:40 to 14:20, posture compliant.

She had badged in. Of course she had badged in; it was a

work event. Her phone had joined the venue Wi-Fi, of course it had; the ballroom signal was poor. Each record was true. Each record was hers. Together, with the charge, they composed something that had never happened: an expense-approved meeting between a Traceframe engineer and a foreign consultant, documented end to end, on the day that an Alero verification window opened in Neuquén at 13:30 local time and closed at 14:00.

She checked the timing against what she had, which was not much: one closed incident, three early receipts and a habit of reading other people's logs. But if a test had run in that window, then every record around her lunch was not surveillance. It was custody. Someone had built her an alibi for a sale she had never witnessed, using nothing but her ordinary Friday.

At home she laid it out on paper, actual paper, the way she did when she needed to see who knew what. The card number lived in four places: her wallet, Traceframe's expense provider, the bank and whatever merchants kept tokens. The badge lived in Traceframe's facilities system. The device signal lived in the security stack. No single party saw all four. Whoever assembled the picture had either compromised four systems or been handed four pieces by people who each thought they were doing something routine.

That was when she caught herself doing the other thing, the thing she would think about later at two in the morning with genuine shame. She was ranking. If I had to un-freeze one card tonight, which one? The personal, because the Subte needs it. And if I had to rebuild the expense trail, where would I have said lunch happened? Not the hotel. The café on the plaza, the one with the bad Wi-Fi, because I always choose the café with bad Wi-Fi when I want to be left alone with a

document.

She stopped writing.

Those were not decisions a stranger could audit. Those were habits, worn down to grooves, visible only to someone who had watched her make a hundred small choices about money and routes and tables. Mateo used to read them out loud at the kitchen table, amused, merciless: *you froze the wrong card first, you always freeze the Subte card last, you chose the café before you chose the excuse*. Two years gone and she still ran his inventory of her like a checklist every time she moved money.

Except Mateo was in Córdoba, or Montevideo, or wherever volume found him. And Noa Ben-Ami had left the country on the evening flight, or so her badge said, or so somebody's log said.

Sofía looked at her page of notes and understood the shape of the thing. The purpose of the lunch had not been recruitment. Recruitment had been the conversation she was meant to leave with. The purpose was the records. She had walked into a room and generated, just by being there in the ordinary way, custody-grade evidence placing her beside Vesper at the exact hour a zero-day test certified itself in another province. If that sale ever needed a human face, hers was now attached, truthfully, verifiably, with her own badge and her own phone.

Her financial mess made it better. That was the part she made herself write down, in small letters, because it was the load-bearing fact. An SRE with clean books would call a lawyer and laugh. An SRE with an undeclared San Marino account would hesitate, and hesitation was all the fabrication needed. Her past was not just exposure. It was adhesive. Every true thing she had hidden made the false things stick.

At 23:40 her phone lit the ceiling. An SMS from the bank, automated:

Authorization attempt declined. Card ending 4417. Amount: 4,850 ARS. Merchant: Estación de Servicio, Ruta Nacional 22, Neuquén.

A fuel stop, nine hundred kilometres away, on a card she had frozen six hours before. Somebody was testing whether the freeze held, or whether she would un-freeze it, the way anyone would eventually un-freeze a card, at a petrol station, late, far from home.

She put the phone face down on the nightstand and did not touch it again.

In the morning the declined attempt would still be there, one more true record in a file of true records, composing a woman who had been in two provinces at once. Sofía lay in the dark and listened to the building's water pump cycle, and understood that she was no longer investigating a market.

She was being written into one.

Chapter 6: Technical Merit

The message came through an old channel they had never closed, a bug-bounty forum private thread from 2022 with one new line in it: *Saturday, Almagro, the workshop on the corner. I'll be the one being yelled at about capacitors.*

She went anyway. That was the part she examined afterwards, more than anything else he said that day: not the meeting, which was public and ordinary, but her own yes, sent within four minutes.

The workshop was a repair collective two blocks from the avenue, a garage front where a man rebuilt e-bike batteries at a bench by the window and a woman resurrected dead stereos for a queue of people who had been told by three shops to buy new ones. Mateo sat at the back table with a plate of empanadas and his laptop angled away from the room out of habit rather than need. He looked well. He looked, Sofía thought, like a man whose bets had been paying.

"You froze the wrong card," he said, instead of hello.

"Hello, Mateo."

"You froze the corporate first, then the personal. You always protect the Subte card last, even though it's the one you

actually need on a Sunday.” He turned the laptop around. “I know, because I watched you do it forty times at that kitchen table, and because someone ran the same play on you this week and you reacted exactly on script. Which is interesting, because it means either they have your habits, or they have my habits.”

“And you know about this week how?”

“Lattice talks to me.” He said it plainly, no theatre. “You found something at work. Four quiet pods, receipts older than their own environments. It made noise in places noise doesn’t usually reach. I came to tell you what you’re actually looking at, because nobody else will, and because you’ll check it instead of believing me.”

He opened a folder of screenshots, redacted down to structure, and walked her through a finding. A dependency parser, widely deployed, mostly in logistics software. Murmur, running against a fictional twin of its class, had produced a candidate flaw in eleven days: a deserialization path that let crafted input walk past validation when a specific legacy flag was set. Real. Reproducible. Sofia spent ten minutes reading the proof-of-concept transcript with her professional face on while her chest did arithmetic about how many such systems moved parcels around the cities she loved.

“Submitted where?” she asked.

“To the market. And to the vendor, through a broker, because we do that too.” Mateo’s voice stayed level. “Vesper saw the submission format, knew it wasn’t one of their named humans, and pulled the lever they keep at Alero. Derivative. Prior art, they claimed, from their own 2024 package. The receipt got labelled, the label got priced, and the vendor sat on the patch for five months because a derivative finding is a

scheduling problem and an original one is an emergency.” He scrolled to the last screenshot: a changelog, finally shipped, crediting an internal audit. “Five months. Nobody hacked anybody. The market just decided the flaw mattered less because of who its parents were.”

“That’s one story,” Sofía said. “Told by the side that profits from it.”

“It’s one story told with evidence. Check the dates yourself, they’re all public.” He pushed the empanadas towards her. “Here’s the part that isn’t about my side. You’ve met Vesper’s answer now, the woman in Recoleta. Human names on everything. Ask yourself who can afford to sell their work that way. A named expert with custody discipline and legal cover is a luxury good. My cousin at the university writes world-class fuzzing tools and cannot afford a lawyer to read her own licence, and Vesper’s model prices her out of her own findings. Human only means wealthy only. It always did. The doctrine isn’t about safety, Sofía. It’s a guild badge.”

“Your doctrine has a price list too.”

“Everything has a price list. Mine just doesn’t have a velvet rope.” He shrugged, and for a moment he was the man she had loved, the one who could make an unjust system sound like a door left open. “Reproducible output is true whether a person or a model typed the first line. Judge the result. Everything else is ancestry worship.”

Before she could answer, the door opened and the room’s traffic changed. A young woman came in carrying a cardboard box like it contained a sleeping animal, which in a sense it did: a ruggedized tablet, a cash box and a tangle of cabling that held together the dispatch system for a delivery operation. Behind her, a teenage rider waited with the particular patience

of someone not being paid while he waited.

"Micaela," the battery man called out. "Tu chico está en la mesa del fondo."

Micaela Ferreyra was twenty-seven and built like someone who had been doing four jobs since she was nineteen. She greeted Mateo with the familiarity of a client, put the box on the back bench, and explained the problem to the room in the flat tone of someone reciting weather. The compliance module had died again. The city had changed a rule about invoice codes in March; her tool, which she had built herself, prompt by prompt, over one frantic summer when paper ledgers stopped passing inspection, handled the old codes fine and fell over on the new ones. Someone from the workshop had patched it in February. Now the patch and the update fought each other every second Friday, and today the whole dispatch floor was running on a whiteboard and two phones.

The rider behind her was called Ariel. Micaela introduced him only because he had been waiting long enough to become part of the meeting. He was nineteen, studying refrigeration at night and losing money by the minute while the tablet sat open. Before Micaela built the tool, he had worked for a larger platform whose ranking decided when he saw an order and never explained why. Three missed acceptances could erase a weekend's work. Micaela's operation paid less on the best nights and more on the worst ones, because the riders could see the queue and argue with a person.

"The whiteboard works," Ariel said. "It just works for eight riders. At twelve we start sending two people to the same pharmacy. At fifteen somebody's medicine goes to Caballito."

Micaela's generated tool had not replaced a well-funded engineering team. It had replaced a platform nobody in the room

could appeal to, using rules none of them could inspect. For one summer it had given a small operator the ability to run a queue on her own terms. The technical debt arrived later. The independence had arrived first, and was real.

"Nobody here wrote that component," the stereo woman said, opening the tablet. "It arrived finished. There's a commit history of one author and the author is a word, not a person."

"I can explain what it does," Micaela said, defensive before anyone accused her. "Mostly. When it works."

"Mostly is doing a lot of work in that sentence," Mateo said kindly, and got up to look, because Mateo could not leave a broken machine alone any more than Sofía could leave a bad timeline.

Sofía stayed at the table and watched, because watching was her job.

Micaela knew exactly what the tool was worth. She talked about delivery windows and margins and which rule changes moved money with the fluency of a woman who had survived three governments' worth of paperwork on two wheels and a phone. And standing next to the open tablet, waiting while a volunteer coaxed her business back to life, she went slightly rigid, the way people do beside their own blind spot. She approved the fix when it was offered. She did not ask what it changed. She signed nothing, because there was nothing to sign, and everyone in the room understood the deal being made: the workshop kept her alive, and nobody mentioned that the component at the centre of her company was a thing none of them could explain, including, especially, her.

Mateo saw raw material when he looked at her. She could see him see it: capability arriving faster than institutions, proof that the old gatekeepers were optional. Later, walking her out,

he said as much. "That's who the provenance cartel locks out. She's better at operations than half their named experts. Under our model, her evaluation work would be worth money."

Under their model, Sofía thought, her name would be worth money too, attached to whatever needed a signature. She said none of it. But she took the folded note Micaela left with the workshop, the one with the component's name on it, and she read it twice, because she recognized the shape of the problem from the inside.

Sofía could verify an effect without owning a use. Micaela could neither verify nor own either. Between those two sentences lay the entire argument everyone was having with such confidence, and neither faction's name was on either of them.

At the door, Mateo held out his hand like a businessman closing a meeting.

"One more thing, free of charge," he said. "Whoever is building your alibi, don't chase the records. Records are their home turf. Chase the timing. You of all people know that clocks are the one thing they have to keep honest, because the whole product is *when*."

"And if the clocks are theirs too?"

His smile faded by a degree, the first honest thing to cross his face all afternoon.

"Then you're in worse trouble than I thought," he said, "and you should call me before you call anyone."

Chapter 7: The Maintainer

The handover call was set for Saturday at 18:00 Buenos Aires time, because Martin believed that reliability was a schedule and not a mood. Eleven years of working together and he had missed four calls: two for flights, one for a hospital, one for the blizzard that had taken out half of Toronto's grid in 2023.

At 18:20 Sofía closed the video window and opened his public trail instead, because when Martin went missing from a schedule he was always, somewhere, visible in public.

She found him fast, and what she found rearranged her picture of him.

The library was called Vouch. He had maintained it for six years: a small, serious tool for signing and verifying software artefacts, the kind of project with three thousand dependents and two maintainers, one of whom was now answering issue trackers in prose that read like a man shouting into weather. She scrolled back through the spring. Generated security reports had been arriving against Vouch all year, dozens a week, each one polished, formatted, plausible, and wrong in ways that cost an hour to disprove. In March a contributor had merged one by mistake. In April a real vulnerability re-

port had sat unread for nine days under forty pages of model-written lookalikes.

Sofia opened the April report.

Its author was Nadia Ortiz, a junior developer maintaining appointment software for three neighbourhood clinics. She disclosed assistance in the first line. She had not understood Vouch's signing internals when a damaged appointment export began failing verification; she had shown the failure to a model, asked it to reduce the input and then reproduced the reduced case by hand. The first explanation the model proposed was wrong. The test case was not.

Martin had closed the report in four minutes.

Generated analysis of unsupported input. No demonstrated boundary failure.

Nine days later the other maintainer reopened it. Nadia answered every question she could and wrote *I don't know* beside two she could not. Between them they established that Vouch trusted a piece of package metadata after checking the package but before checking whether that metadata belonged to it. Nothing in the thread could be used as an attack recipe. What mattered was the social sequence: a newcomer with an assistant had found a real discrepancy, the senior maintainer had recognized the shape of forty bad reports and discarded the forty-first, and the project had been less safe for nine days because experience had become a filter too aggressive to admit surprise.

Martin's apology was still there.

You brought evidence. I classified your prose. That was my failure.

Nadia had replied: *The prose wasn't mine. The reproduction was.*

The exchange had been quoted by both camps afterwards. The enthusiasts used it to prove maintainers were obsolete. The sceptics used the forty preceding reports to prove Nadia's success was unaffordable. Nobody quoted the line underneath, from Vouch's second maintainer, saying she had taken a month away because triage now consumed the hours in which she used to write code.

And in May, Martin had written the message that got quoted back at him for weeks afterwards:

Every generated report costs me an hour I do not have. You are not automating security review. You are automating the exhaustion of the people who do it for free. If your tool finds one real bug a year, congratulations, you have priced a volunteer's Saturday at zero to buy yourself a benchmark point.

Arguments followed. Some called him a gatekeeper. Some called him the only honest man in the thread. A foundation mediator offered to help triage. The reports kept coming. In June his commits slowed. In July they stopped, except for one, a documentation change so small it looked like a goodbye: he had corrected the word *principally* to *primarily* in the README, which was the kind of edit a man makes when his hands want to touch the project once more before he puts it down.

Sofía sat with that for a while. She had known Martin as the clock man, the handover man, the colleague who remembered her mother's birthday because it was in the rotation file he kept for the team's on-call morale. She had not known about the Saturdays. Everyone in her profession knew the shape of that argument from the inside; she had simply never seen Martin on the losing end of one before.

At 21:14 her phone buzzed with a message sent through the incident platform's side channel, the one they used for things

that should not live in company systems. It had been composed at 16:02 Toronto time and scheduled for delivery at this hour, which meant Martin had planned its arrival down to the minute before whatever happened to him happened.

Sofia. If you are reading this, I am not reachable, and you have already noticed the pods. Handover notes follow. Read them before the attachments.

1. The regional audit cache is clean. I built it paranoid and it has stayed paranoid. Whatever else is compromised, the timing record is not. That is why it matters, and why people will start asking where it lives. Do not tell them. Not either side. Especially not both.

2. The receipt format is mine. Not literally, but the skeleton. In 2023 I took contract verification work through an intermediary. Defensive testing, I was told, and I believed it because I wanted to believe that being the human who signs was worth something. Attachment one is the invoice. Look at who the payer is.

3. When I understood what the receipts were settling, I did what clever men do instead of wise ones. I made a record of my own. Both factions would pay for it, which means both factions will come for it, which means I thought that made me safe. Two buyers do not create independence. They create an auction. I understand that now.

4. If I go quiet, do not rescue me. Validate nothing of mine until you can prove the interval yourself. Your clocks are good. Trust the boring ones.

Do not let either side own the time.

Two attachments. The first was an invoice, photographed flat on wood grain: consulting services, verification and attestation, thirty hours monthly, issued by a shell with a registration number in a country she associated with ship registries. The payer field said **Alero S.A.**, and the payment reference said **witness-04**, and there was Martin's legal name, attached

to the market she had found inside her own cluster, going back further than her incident, further than her anomaly, into the years when he had been teaching her how to read clocks honestly.

The second attachment was encrypted, keyed to a passphrase she recognized: the name of the first outage they had ever worked together, a BGP blunder in 2019 that had taken down half of Patagonia for ninety minutes while everyone blamed the cables. She typed it in. The file decrypted to a single page of hashes with timestamps, no explanations, titled only:

mixed continuity record: interval draft 3

She did not know yet what it mixed. She wrote down its hash anyway, on paper, in the notebook that lived in the kitchen drawer, next to the one fact she now held about the world: Martin Vale, the most honest man in her professional life, had been on Alero's payroll as a human witness, and had tried to sell that fact in both directions, and had vanished between the bids.

Outside, Saturday night came up over Palermo the way it always did, doors and music and a scooter with a bad exhaust doing wheelies on the corner. Sofía closed the notebook and put the kettle on, because her hands wanted procedure and there was no procedure left in the queue.

Two days ago she had found four pods certifying the future. Tonight she knew the future had witnesses, that witnesses could be bought by the hour, and that the only independent clock left in the story was a cache in São Paulo that a missing man had built to be annoying to forge.

Whatever came next would arrive asking for it.

Part Two: Two True Histories

Buenos Aires, Rosario, Paraná. Ten days before settlement.

Chapter 8: The Wallet Years

On Sunday morning, with the city hung over and quiet, Sofía did what Martin’s message had told her to do in spirit if not in letter: she audited herself before anyone else could.

The notebook came out of the kitchen drawer, along with the second notebook, the older one, whose pages had gone soft at the corners from a winter of being carried in a coat pocket. She had kept records even then. That was the joke she never made out loud: the woman who hid money from the tax office kept better books on herself than most companies keep on their fleets.

She rebuilt the timeline year by year, writing balances as ranges instead of numbers, because ranges were honest about what memory could support.

2022. Remote work in dollars while the peso burned. The gap between what her invoice said and what arrived in her Argentine account was a tax on geography: withholding, conversion at the official rate, card spending surcharged into absurdity. She remembered standing in a supermarket queue doing division in her head to decide whether

the yerba was cheaper this week or last, and the answer changing between the queue and the till.

2023. Inflation at numbers that stopped feeling like weather and started feeling like altitude. Two hundred and eleven per cent across the year; the month she got married to no one and told no one anything, December, prices had risen a quarter of themselves in thirty days. Her dollar salary sat in an Argentine bank melting like an ice sculpture at a demonstration. And Mateo, who had been reading her statements before she read them, because that was how they were together then, put his phone on the table one night and said:

“Your next payment lands on Thursday. By Friday it will be worth ten per cent less than the invoice says. By Christmas, half.” He turned the phone towards her. On it was a stablecoin order screen, dollars wearing a blockchain’s clothes. “Convert the day it lands. It holds its value. You can change it back when you need pesos, in pieces, when the rate isn’t insulting.”

She remembered the specific quality of that moment, because it had not felt like risk. It had felt like being handed an umbrella in rain she had stopped noticing. Relief first, arithmetic later. The fee was small. The transfer took four minutes. Her money, for the first time in two years, was where she had put it an hour ago.

That was the first conversion. The notebooks recorded it as 400 USDT at 231.80, which was either careful bookkeeping or superstition, and neither of them knew yet that they were founding something.

Because the route became a habit, and the habit became infrastructure. A payment processor in Montevideo. An

account at Banca Rocca, opened on Mateo's advice through an intermediary he trusted, because San Marino had signed nothing and asked little. Then Proofbox started paying in Bitcoin, and there were spreadsheets with two columns marked *declared* and *actual*, and the gap between them grew the way gaps do, gradually and then enormously. She underreported the foreign account. She underreported some gains. Each individual decision had been small, defensible, survivable, the financial version of signing off a cleanup clause you had not fully verified, and the sum of them was a woman who could be ended by a spreadsheet.

She wrote the last balance in the range format and closed the old notebook.

The new message arrived at 14:20, through the bug-bounty forum thread, the same dead channel Mateo had used on Friday. Not from Mateo's account. From an account created yesterday, named with a string of characters that meant nothing and would trace to nothing.

No greeting. One line of text and one image.

We mapped this one. Ask your ex how many more we hold.

The image was a screenshot of a block explorer showing a single transaction from March 2023: her first Bitcoin movement, the proceeds of the first Proofbox job, travelling from the wallet Mateo had set up for her to the consolidation wallet they had shared. An amount, a date, two addresses, a fee. Public data, technically. Every chain analysis firm in the world could have found it.

But she had never used that wallet anywhere near her name. The exchange account was in a jurisdiction that did not answer foreign requests quickly. The address had appeared in no leak she knew of. For someone to hand it back to her, attached to

her, they had linked the unknowable to the undeniable: they had connected a pseudonymous address to Sofía Alvarez, and chosen a transaction intimate enough to prove it.

Not all of it. One transaction. The way you show a hostage-taker that you know where the house is by mailing back one photograph of the door.

She looked at the date again. March 2023. The week of the first conversion, the umbrella in the rain, Mateo's phone flat on the table between their plates. He had proposed the route. She had experienced it as rescue. The notebook said so, in her own handwriting, and now a stranger's message said it too, in the grammar of leverage: *ask your ex*.

She understood the instruction perfectly. They wanted her to believe the map of her history lived in Mateo's head, so that every fear she felt would point at him, and every question she had would route through him, and the faction that owned him would own the channel to her.

It might even be true. That was the elegant cruelty of it. Some of what she was seeing was certainly Lattice's data work, clusters and heuristics and probabilistic matching run at scale against every Argentine wallet that moved in the crisis years. And some of it might simply be Mateo's memory, accurate as a knife drawer, doing what it always did.

She could not tell which. Neither could they, perhaps. Neither could he.

Sofía photographed the message, printed the screenshot, and filed it behind Martin's invoice in the kitchen drawer, building the file she hoped she would never need: evidence of who already knew what she had spent three years hiding.

Then she updated the timeline in the new notebook, one line, dated today:

My past is now a weapon that at least three people are aiming.

Chapter 9: Fork

Martin's public trail ended at a dead mirror.

She had spent Sunday afternoon following him through the commons the way you follow anyone through a city at night, by what stays lit. His Vouch commits had stopped in July, but his activity on a community forge had not: comment threads, review notes, a long argument about attestation formats that he had won on points. The forge was called La Forja Sur, a volunteer-run instance hosted, according to its footer, on donated hardware somewhere in Buenos Aires. On Thursday, the day her pods first certified the future, the mirror's availability page had gone grey. Since then it had returned nothing but connection timeouts.

The instance administrator's handle resolved to a real name in the way small volunteer infrastructure always eventually does, through a sponsorship page and an old conference talk and a photo of a man patching a network switch while holding a guitar case between his knees.

Julián Ibarra.

Sofía sat back from the laptop and let the name have the room.

They had been together from 2016 to 2019, which meant he knew her from before the inflation years, before Proofbox,

before she had become a person who kept two notebooks. The relationship had ended without betrayal and without a final conversation, just the slow arithmetic of two people whose lives were converging on different answers to the same question. She had been building redundancy into everything: savings, systems, exits. He had been building nothing at all, on purpose, living out of an instability he treated as a design choice rather than a defect. She had mistaken it for immaturity. She was no longer sure. That was the thing about Julián: he was the one person from that decade who had kept every promise she ever watched him make, and none of them had been promises to stay.

She had avoided him since well before Mateo. Avoidance was easy; Buenos Aires was technically twelve million people arranged in practical clusters of two hundred.

The forge's status page listed its admin contact, and the admin contact's other project was the sound system at a club in Palermo called Cable Rojo, which posted its open mic lineup on Thursdays.

He was going to be ten minutes from her apartment tonight, singing.

Cable Rojo occupied the ground floor of an old printing house, narrow neon over the door, graffiti layered on the side wall like sediment, a bar in front and a live room behind where local bands played to people who owned their amplifiers. The open mic ran early Sundays. Sofía paid the cover, took a position near the back pillar with a beer she did not want, and spotted him immediately, because some things do not change: Julián on a stage looked like a man standing exactly where his life had put him.

He was midway through *Livin' on a Prayer*.

Of course he was. Every open mic in every city on earth held one man singing that song, and in this one he was good at it, guitar tight against the backbeat, eyes shut on the high part, the crowd of regulars giving him the chorus the way crowds give choruses, as rent. Sofía stood at the pillar and watched her ex-boyfriend sing karaoke-adjacent rock with total professional commitment, and felt the strange vertigo of timelines crossing: the woman who used to wait for him after sets in another decade, holding two notebooks and a federal crime, waiting for him again.

She arrived too late to catch him before the set. There was no interrupting this without becoming the room's problem; the song had a contract with its audience, and she was not going to be the person who broke it. So she waited through the key change and the last chorus and the applause, and watched him see her.

It happened the way it happens in rooms like that: he was scanning the back of the room during the applause, the way performers do, checking the size of what they earned, and found her at the pillar, and went still for half a second before finishing the bow. He lost none of his stage composure. He thanked the crowd, handed the guitar to the next act with a cable-coiling ritual so practiced it looked like sleight of hand, and came over unhurried, wiping his hands on his jeans.

"Sofía Alvarez," he said. "In my bar."

"Your bar has a hole in the ceiling."

"The hole is load-bearing." He looked at her properly then, and whatever he saw made him skip the social part entirely, because Julián had always been able to read the difference between a woman coming to say hello and a woman coming to ask for something. "You didn't come for the music."

"I didn't know you'd be playing. I need to talk to you about La Forja Sur."

"Of course you do." No bitterness in it, just recognition, and something else she had forgotten: patience. "Not here. Not over that noise." He glanced at the stage, where the next act was tuning badly. "Give me twenty minutes. I have to hand over the desk, the mixer and a very temperamental compressor to Nacho, and if I abandon them mid-handover I will pay for it all month."

"You're doing a handover at your own open mic."

"Everything is someone's production," Julián said, "if you intend to be allowed back."

She watched him do it from the bar while she paid for the beer she had not drunk. It was, she had to admit, its own kind of competence: cables labelled in handwriting, a checklist taped inside the desk lid, each item pointed at and confirmed aloud, Nacho nodding along, the compressor coaxed with two knobs and a threat. Nobody had ever run an incident call better. He finished, dusted his hands, collected a canvas bag that clinked, and walked her out into the night air, where the graffiti wall glowed under the neon and the winter-damp street smelled of fried onions from the parrilla two doors down.

"There's a bodegón on the corner that stays open late," he said. "I haven't eaten, and I'm not discussing a broken machine until I have."

The place was all fluorescent light and football on muted television, six tables, a waiter who greeted Julián by name. Milanesa napolitana for two, he ordered, before she could open her mouth, plus a bottle of the cheap red that bodegones are legally required to serve in unlabeled bottles.

"Julián."

"Eat first. Then tell me who died."

"Nobody died." She cut into the milanesa when it came, thin beef, ham, sauce, a slab of melted cheese over all of it, engineering rather than cuisine, and understood why he had ordered it: this food takes twenty minutes to eat no matter what you say. The table could not be cleared around her. There would be no quick version of tonight.

So she told him. Not everything; she told him the shape. A colleague who had gone missing after finding something wrong in the systems they both cared about. A public archive, La Forja Sur among them, that had stopped answering the same week. Evidence that mattered, sitting on failing hardware, in the custody of an administrator whose name she had known since before any of it.

Julián ate through most of it, listening the way he listened, which was completely and without performing it. When she ran down, he wiped his mouth and said:

"You've narrowed that confession until it sounds like a work ticket."

She opened her mouth to deny it and closed it again, because he was right, and because he had seen it in 2018: her habit of converting anything raw into a request with acceptance criteria. *Here is the problem, here is the scope, here is what I need.* Confession as service request. Intimacy filed under incident.

"Martin Vale," Julián said. "That's the missing colleague. The Vouch man." He turned his glass by its base. "I don't know anything about any of it. You understand that? I maintain a mirror. People post arguments about software licences at me. Whatever Martin was into, it never touched the forge."

"His last public activity was on your instance. Then your mirror died the same week."

"My mirror died because it got flooded." He said it without heat, an engineer reporting facts. "Three months of generated issue reports, thousands of them, polished garbage, every one formatted beautifully. The storage filled, the search index choked, and the volunteer who used to clean the queues gave up in February because he said it was like shovelling the sea. I've been meaning to rebuild it and I've been meaning to rebuild it." He looked at her, and she watched the decision happen behind his eyes, not quickly, the way his decisions always went, all the way down to foundations before anything moved. "But the data might still be there. Storage dies slower than websites do."

"You'll look?"

"I'll preserve it. Tonight, tomorrow, whenever Nacho doesn't need me." He pointed the fork at her, gently. "And Sofía, I'll do it because that archive is mine to keep. It's the only reason I do any of this. Somebody has to remain after everyone interesting leaves. But I'm not your incident responder, and I'm not getting pulled into whatever this is past the edge of my own machine. One look. One report. That's the whole ticket."

"That's all I'm asking for."

"No," Julián said, signing the bill without looking at it, "it isn't. But it's all I'm granting."

She paid anyway, over his objection, and tapped her card on the terminal with a small grim satisfaction she did not explain to him. Let the record show that on Sunday night she bought dinner in Palermo with her own frozen-and- unfrozen personal card, in full view of whatever was watching, one hon-

est location record in a week that was filling up with dishonest ones.

Outside, the neon hummed and the night had softened. They stood on the corner for a moment in the old geometry, both knowing the moment, neither feeding it.

"He's involved, isn't he," Julián said. Not a question. "Whoever's missing you won't name. The one who taught you the money routes. Your Mateo."

Sofía said nothing.

Her pause answered him, and they both heard it, and Julián nodded slowly, putting his bag over his shoulder like a man who had just learned the price of the favour he had already agreed to do.

"Careful," he said. "You always were worst at the things you were best at."

Chapter 10: Read-Only

The rehearsal room sat behind the club, up a staircase that smelled of solder and old beer, and it held, among amplifiers and a drum kit in a case and posters for nights nobody would remember, a computer rack.

Not a server room. A rack: five machines of different vintages on wooden shelves, cable-tied into stability, a consumer router, an uninterruptible power supply with its battery replaced so recently the sticker was still on. La Forja Sur, in the flesh. The community forge ran here because Julián had bartered hosting against sound engineering, which meant the archive lived above a rock club, cooled by a fan that had opinions.

“Three nodes run the cluster,” Julián said, kneeling at the shelf. “This one is storage. Or it was.” He laid the back of his hand against the case, the same gesture Lucia Barrenechea would have recognized four hundred kilometres south. “It’s warm, which means it’s alive, or it means it’s dying loudly.”

“You didn’t check it since Thursday?”

“I checked that it pinged.” He pulled the machine out with both hands, cradling it like the drum case. “That’s the honest confession of every volunteer admin alive. You monitor what you have time to mourn.”

They put it on the workbench under the good lamp. He opened the case while she fetched the screwdrivers he named without looking up, and inside, the smell of hot electronics rose off two of the four drives like a smell of burnt toast from a kitchen where something had already happened.

"Two drives down, same pool," he said. "Raid protects you from one failure. Nothing protects you from one failure plus neglect." He connected the drives one at a time to a recovery dock, listening to each spin-up with his head tilted like a doctor with a stethoscope. The first drive chattered, caught, and presented a file system. The second made a sound like a card in bicycle spokes and gave up.

"Pool's degraded but mounted," he said. "We get one pass at this, read-only, no repairs, no cleverness. Copy first, look second. Where's your external?"

She had brought two, because she was who she was.

While the first image crawled across at USB speed, Julián walked her through what had actually happened to the mirror, and it was a story she recognized from her own week wearing different clothes. In February, La Forja Sur had appeared on some list somewhere, and the generated issue reports had arrived: hundreds a day, then thousands, each one grammatical, formatted, confident, and useless. The volunteer triager quit. Queues filled. And underneath, the storage pool crept towards full, because every polished nothing still occupied real blocks.

"The site didn't get hacked," Julián said. "It got loved to death by machines. Storage hit ninety-nine per cent Thursday morning, the database froze mid-write rather than corrupt anything, God praise its little rules, and the whole thing fell over looking exactly like an attack if you only glance at it." He

glanced at her. “Which is why you’re really here. Your missing man’s last messages are in this thing, and this week everyone wants it to have died dramatically.”

“Would you have looked without me?”

“No. I’d have rebuilt it empty next month and hated myself quietly forever.” The copy progress bar moved. “There. Now we look properly.”

The surviving persistent volume held the forum data: users, threads and the append-only event log of everything the application had ever done. Julián gave her the events and kept the disks, an easy division of labour that they fell into without discussing, because it matched their shapes. He read hardware. She read history.

She reconstructed the chronology outward from Thursday’s crash, and within an hour had the answer to the question she had carried across town.

Martin’s account had logged in on Wednesday at 21:41. He had read a thread. At 22:03, the thread vanished from the index. Not deleted in any way the application admitted to: no deletion event, no moderator action, no audit trace. The discussion simply stopped being listed, and its direct URL started returning not-found, while the underlying posts sat untouched in storage the whole time.

“Removed at the application layer,” Sofía said slowly, “after the mirror had already stored it. Someone with application access hid a thread without using any feature that admits hiding things.”

“So not clock skew,” Julián said dryly, “and not my dead drives. Somebody walked into the house without breaking the lock.” He did not ask how that was possible. He just wrote *app-layer compromise* on a sticky note and stuck it to the monitor,

because some facts you file before you feel them.

The thread itself was recoverable from the raw store, and she read it twice while Julián pretended to be busy labelling cables.

It was Martin at his best and worst. The subject was a fork war in a free-software project that maintained a dependency scanner. One branch of the community had adopted a hard rule: no AI-origin reports accepted, full stop, protect the commons, protect maintainer attention. The other branch held that tools were tools: assistance allowed, disclosed, reviewed, signed by a human who understood it. The argument had run for months, civil by forge standards, which is to say there were survivors.

Into it, eleven days before he disappeared, Martin had posted a technical observation that stopped the argument cold. A zero-day claim currently being traded, he said, in a market he did not name, had appeared on both sides of the fork. On the no-AI branch it carried one authorship trail. On the disclosure branch it carried a different one. Same claim, same proof-of- concept structure, two histories, and the difference between them was worth, he suspected, real money to whoever was brokering it. Somewhere between the branches, the claim had changed parents.

Authorship is supposed to be the one thing a fork cannot lose, he wrote. You can fork code and both copies stay true. Fork a fact's provenance and at least one copy is a lie. I am going to find out which.

Nine days later, the thread was hidden. Four days after that, her pods certified their first impossible receipt.

"There's your missing discussion," Julián said quietly, reading over her shoulder, because of course he had read it

too; keeping archives makes curious people. “And there’s why someone reached into my machine to bury it.” He straightened up. “One problem. This thread references a dependency: the scanner both branches use for report triage. It’s mentioned by name eleven times as the thing that got flooded and broke.” He pulled up the package page on his phone. “Know who maintains the biggest deployment of that scanner in this city? The delivery co-op girl. Micaela. My guys patched her instance in February when the rule change melted it. Same component, same generated core nobody can explain. It’s the hinge both sides of this fork swing on, and it lives in a shop tablet in Balvanera.”

The chain assembled itself in Sofía’s head with an almost audible click. The flood that killed the mirror and the patch that saved Micaela’s business were the same component wearing two fates. Vesper’s people wanted Martin’s thread buried. Lattice’s people wanted the claim’s second parentage never traced. And sitting underneath both, holding the whole structure an inch above the ground, was a piece of software whose own author could not explain it.

“Call her,” Sofía said.

Julián checked the time. “It’s one in the morning.”

“Her dispatch floor runs until two on Sundays. She told us that yesterday.”

He made the call from his own phone and put it on speaker only after Micaela agreed. Behind her voice came motorbikes and the double ring of the shop door.

“The February backup,” Julián said. “Do you still have it?”

“On the workshop drive, unless somebody cleaned it. Why?”

“The generated component in your compliance tool shares

a package with the scanner that drowned our forge. I need permission to compare the copy we made with the public package history. Read-only. No customer data."

Micaela was quiet.

"You said the repair was done," she said.

"It was. This is not a repair."

"Then what am I agreeing to?"

It was the right question and neither engineer answered quickly enough.

Sofía moved closer to the phone. "We're trying to establish when that component entered your tool and which parts you changed afterwards. Not whether you wrote it. Not whether using it was wrong. Only the history."

"And if the history is bad?"

"It stays true."

Micaela laughed once, without humour. "That's a person-with-salary answer. If the history is bad, twelve riders don't get paid on Tuesday."

"Then say no."

The offer stopped her. People had been calling Micaela raw material all week, even when they praised her. Raw material did not get a refusal.

"Read-only," she said. "Julián keeps the copy. You send me what you conclude before you send it anywhere else. And if you don't understand something, you write that you don't understand it. Don't turn guessing into my history."

"Agreed," Sofía said.

"One more thing. That component wasn't all rubbish. It kept invoice matching alive through the March rule change. I know nobody likes hearing that after it breaks. It still did."

"We'll write that too."

Micaela hung up to release a rider whose phone had died. Julián wrote her conditions on a label and stuck it to the backup drive before touching anything else.

Julián unplugged the recovery dock and squared the drives on the bench like tools after a job.

"Here's my limit, and then I have to catch the last bus," he said. "I'll keep this image safe. Cold copy, offline, two places. And I'll do one comparison for you when you need it, properly, with time to do it right: whatever timeline you build, I'll check it against the physical record. Once. After that, I'm the guy who keeps a mirror, nothing else." He looked at her, and asked nothing about Mateo, nothing about the money he could smell behind all of it, nothing about why her hands were not steady on the screwdriver tray. "You used to leave rooms without saying goodbye. That's the only thing I ever held against you. Say goodbye tonight."

"Thank you for the mirror."

"That's not a goodbye."

"Goodbye, Julián." It came out steadier than she expected. "Thank you for keeping things."

He nodded once, released her from eye contact with visible mercy, and turned back to his rack.

On the workbench the recovered volume sat mounted read-only, its contents now copied to two drives, one of which went into her bag and one into his fire safe. An archive that had been loved to death by machines had given up one true sentence before dying: a fact can be forked, and somebody had paid to hide which copy lied.

Outside, the club had emptied. The neon over Cable Rojo buzzed and flickered, holding its colour the way the drives held their data: imperfectly, warmly, against all advice.

Chapter 11: Assisted By

On Monday she went back to the fork.

Martin's recovered thread gave her the map. The territory was two issue trackers, public, mirrored, one for each branch of the scanner project since the community split. Somewhere between them, a claim worth real money had changed parents. She booked a meeting room nobody wanted, put the two histories side by side on the largest screen she could find, and started reading the way she read clocks: for disagreement.

The claim itself stayed sealed behind the usual doors, and she did not try the doors. What travelled in the open was the paperwork. Reports, reproductions, maintainer correspondence, credit lines. Provenance leaves records even when exploits do not.

On the no-AI branch, the flaw had been filed by one researcher with eleven years of history, a distinctive prose style and the small inconsistencies that living writers carry: version numbers remembered wrong once and corrected later, a hypothesis abandoned halfway through and replaced by a better one. A human trail. Messy in exactly the way truth is messy.

On the disclosure branch, the same flaw wore different clothes. Its report had been amended twice, and the amendments carried fresh names: researchers she half knew from

conference programmes and bounty leaderboards, five of them in total across the record, each credited with confirming the finding under disclosed assistance, each with a plausible public face.

She diffed the confirmation texts, and the room got colder.

All five confirmations described the trigger condition the same way, and all five described it wrongly. Each stated that the parser failed open when a specific legacy flag was unset. Anyone who had actually run the working proof-of-concept would have found the flag irrelevant; the real path went through the deserializer's fallback branch, which the flag merely made louder. It was the kind of mistake a reader makes from a summary. It was not the kind of mistake a person makes after reproducing a thing with their own hands.

And it was the same mistake five times. Identical in wording, down to the bracketed reproduction counts and the same odd habit of calling the legacy flag *the old switch*. Independent humans diverge. That is what independence means. Five people making the same mistake is one author holding five pens.

She photographed the diff, filed it, and sent the five handles to the dead bug-bounty thread, because she had promised herself she would stop routing questions through Mateo and had already broken the promise.

His reply came in nine minutes.

All five reproduced it. Signed confirmations, timestamps, their own environments. Every claim in that queue runs green before it ships. You of all people should respect that. Judge the result.

Who wrote the result, she sent back.

The result is the result, he wrote. Reproduction is the only priesthood I recognize.

But reproduction, she thought, was not the same as under-

standing, and the public record agreed with her. Eight months ago, a branch maintainer had pressed two of the five credited researchers in an open thread, politely, asking each to walk through the decisive step. Both had eventually replied, from different countries, six weeks apart, and their answers matched almost word for word: the flaw was load-dependent and best demonstrated under sustained traffic on affected versions, which was a sentence shaped like an explanation and contained none. Neither mentioned the fallback branch. One of them had then stopped answering. The other thanked everyone for a great discussion.

She wrote one line in the notebook: *Reproduced is not the same as owned*. Underneath it, because honesty had a cost she was trying to stop deferring, she added: *Ask me about midnight cleanup*. Then she closed the notebook before the sentence could grow friends.

At nine-thirty she took the incident file home and spread it across the kitchen table. Four receipts from her pods, printed Thursday night in the office because printers were the one system nobody compromised on purpose. Compact, signed, boring. She reread them line by line, looking for anything she had missed while the incident channel owned her attention, and stopped at the attestation clause.

The witness attests that the environment named herein remained sealed for the stated interval, and that the recorded output was reproduced therein. This receipt attests the interval, not the outcome.

She knew the sentence. Not ones like it. That one.

The kitchen table in 2023 had been smaller, in the apartment she had shared with Mateo, and the pizza box on it had been balancing on the second notebook, which was already becoming what it was. Their first paid verification job had arrived

through a broker neither of them ever named: a client who suspected a rival lab of faking reproductions and would pay for proof, sealed and witnessed, that a claimed result could be regenerated in a clean environment. Mateo had drafted the engagement letter in an evening, full of confident verbs. The witness guarantees. The witness certifies. The witness warrants the integrity of the finding.

The technical work had taken longer. In those years assistance still arrived as fragments: a completion inside the editor, a function in a chat window that somebody copied into the repository, an explanation you distrusted until the test agreed. Mateo asked a model to write the reconciliation tool that compared the sealed environment with the client's manifest. It produced something plausible in forty seconds and wrong in three ordinary ways. It sorted version numbers as text, treated missing fields as empty ones and assumed every clock included a zone.

They fixed it together at the table. Mateo supplied examples. Sofia wrote the acceptance cases. Each correction went back through the chat and returned as a new block they reviewed line by line before pasting. By two in the morning the tool had reduced six thousand manifest entries to eleven disagreements, ten of which were harmless. The eleventh was the client's whole case.

Without the assistant, they would have spent two days writing glue and billed one. With it, they finished before breakfast and understood every line that survived. Sofia remembered the relief more clearly than the caution. The machine had not acted like an author or an employee. It had acted like a fast junior who forgot everything between questions and never minded being corrected. She had thought: *this is what it is for*.

Three years later Mateo's agents read whole repositories, changed several files, ran their own tests and retried until an evaluator turned green. The human glue work had disappeared. So had the long evening in which two people noticed what the tool assumed.

"You're promising things we can't deliver," she had said, reading it with a pen. "We can't warrant the finding. We've never seen the finding. We can warrant one thing only: that the environment stayed shut and the output came out of it during a window we can defend."

"So the receipt says nothing?"

"The receipt says exactly enough." She had crossed out his sentences one at a time, feeling the strange pleasure of it, subtraction as craftsmanship. "It attests the interval, not the outcome. If anyone ever asks us what we verified, the honest answer is on the paper. Nobody can pressure us into signing more than we know, because the language won't hold it."

He had looked at the narrowed clause for a long moment, then smiled in the way he did when someone handed him money. "You just invented a product," he said. And in a sense she had. Proofbox's clients paid, then brokers paid, then a clearinghouse in Patagonia bought the process wholesale, right down to the wording, and neither of them had thought about it in years.

Sofía sat in her own kitchen with the printout flat under the lamp and felt the reinterpretation move through her like weather. She had told herself the old story for three years: that she had operated Mateo's system, signed what he built, been the careful half of someone else's ambition. The paper said otherwise. The grammar every witness receipt in that market spoke, the exact hinge that let a human name attach

itself honestly to output it never understood, was a sentence she had written to protect herself at thirty. She had built the narrowest possible container for responsibility, and the market had industrialized it. Lattice added authors the way she had once added clauses: precisely, defensibly, each addition small enough to survive being read alone.

She had thought she was writing a shield. She had also written the stencil.

Her phone buzzed against the table.

Traceframe identity service. Routine notification, the kind she had configured herself, the kind that usually caught someone enrolling a tablet for the on-call rota.

A new device has been attested to your profile.

She opened the entry. A laptop, macOS, hardware fingerprint freshly minted, hostname `sofia-alvarez-mbp`, one character different from her actual machine's name. Enrolled Sunday evening at 21:47, authenticated cleanly against the corporate identity provider, granted trusted-device status. No failed attempts preceding it. No alarm raised anywhere, because the enrolment had used credentials that were, as far as the directory knew, hers, presented by a machine whose attestation was valid, at a time when she had been standing at the back of Cable Rojo watching a man she used to love finish his open-mic set.

Not stolen, she understood, reading it twice. Assembled. Someone was building a Sofía out of true parts, a device that had genuinely passed attestation, a profile that had genuinely been trusted, a woman who existed in every record except the one that mattered: the chair she had been sitting in.

She ran the revocation procedure with steady hands. Sessions killed, password rotated, hardware token cycled, ticket

filed to the identity team with the facts and none of the theories. The screens confirmed each step in green.

The attestation stayed.

Revocation worked forward, never backward. The device could be untrusted from this minute on, but the record of its enrolment was a historical fact now, filed and timestamped, one true entry in the growing archive of a woman who had never sat in that chair. You could revoke a device. You could not revoke having trusted it.

She turned the laptop screen towards the dark window, where her own reflection sat at the kitchen table with the incident file, and for the first time that week she could not immediately say which of them the records described.

Chapter 12: Vellum

Noa Ben-Ami kept two rules about Buenos Aires meetings: never the same hotel twice, and never before eleven, because the city did not believe in mornings and pretending otherwise made people honest at the wrong moments.

She chose a café on the quiet side of Recoleta, deep enough from the window that a phone across the street would capture table shapes rather than faces, and arrived twenty minutes early to watch Sofía cross the square. The engineer walked like her timelines, Noa thought. Straight lines, checked corners, no wasted acceleration.

That morning's report from the Buenos Aires counsel had been four pages long. The engineer had traced Lattice's laundering in public records alone: five credited names, one inherited error. Noa had read the summary twice, once as intelligence and once, unwillingly, as admiration. Then she had cleared her morning.

Sofía sat, declined coffee, and placed her phone face down on the table, which Noa noted was either courtesy or theatre.

"You found the five names," Noa said. "I will not waste your time asking how."

"Your turn," Sofía said. "Last Friday you sold me custody. Named humans who understand what they sign. On Sunday

night I watched a machine hide Martin Vale's thread inside an archive it had no business entering. So before we talk about anything else: does Vesper run models or not?"

A fair question, asked without heat, which made it harder. Noa turned her cup a quarter turn on its saucer, an old habit she had stopped apologizing for.

"Yes," she said. "We run one. Its name is Vellum, and you will understand Vesper badly if you leave here thinking that word is a contradiction. May I show you?"

She opened her laptop far enough to share, not far enough to surrender it. Vellum's interface was plain to the point of austerity: queues, confidence scores, a column of provenance classifications. Noa walked Sofía through it the way she would walk a colleague through a radar display, because some of it deserved the comparison.

When Vellum began, Noa had read every hostile case it generated. That was two years ago, when a case arrived as a paragraph and a test condition and the day's work fit inside the day. Then model-originated submissions multiplied, Vellum's operators added autonomous retries and the cases became families whose members changed as rival systems adapted. Noa now read the high-confidence exceptions and a weekly sample. She could explain the sampling policy, the appeal path and the last twelve failures. She could not honestly claim to have seen the whole instrument she continued to call subordinate.

Nobody could. That fact frightened her less than the available alternatives, which was how most professional beliefs survived acceleration.

"Vellum reads the market the way your pods read environments. It fingerprints output that is likely model-generated.

Style alone proves nothing, so it does not stop there. It generates hostile cases against every claim in circulation. Real claims survive hostile cases. Manufactured ones tend to die in predictable places, because a model that invented its finding cannot repair its own reasoning when the case breaks it." She scrolled. "And it drafts dead ends. Plausible research directions that lead nowhere, fed back into channels Lattice's agents are known to crawl. A weapon aimed at automation, not at people. That is the distinction, and I hold it precisely. A tool that challenges a human expert may be used by the expert. An expert of record must be a person. Vellum is an instrument. Nothing it produces carries our name until a person understands it and signs."

"Instrument," Sofía repeated. "It classifies. It fabricates bait. It decides which findings live. At which point in the pipeline does the instrument's judgment get read by a human who could disagree with it?"

"The classification thresholds are set by people."

"That is not reading the judgment. That is signing it in advance."

Noa had heard the objection before, from better defenders than this engineer, and had answered it the same way every time, because the answer happened also to be true. "Every instrument in this market makes judgments. Your cache decides which packets matter. Murmur decides which flaws matter. The question that separates decent operations from corrupt ones is whether a named person can be held responsible for what the instrument does. At Vesper, yes. Always. You have met our paperwork."

"I've met your lunch," Sofía said dryly.

There it was, and earlier than Noa expected. She did not

look away.

“Your corporate card bought a meal it did not eat,” Noa said. “I know. I have complained about it internally with more heat than I will show you, and I will tell you what I told them: manufacturing presence beside my name is a technique I associate with the other side, and someone on ours has decided you are worth the technique. I did not order it. I am not certain who did. Both halves of that sentence should frighten you, and I will not insult you by pretending they comfort me.”

Sofía studied her for a moment, recalculating. Noa let her. Trust built on shared enemies was still trust, and the engineer needed somewhere to stand.

“Show me the harm you claim to prevent,” Sofía said instead. “Not the theory. The bodies.”

So Noa closed the laptop halfway and gave her three, from files already sanitized, because she had chosen them before the meeting. A maintainer of a mapping library who received four hundred generated reports in a season and missed, inside the flood, the one handwritten report that mattered; he had announced his resignation in a commit message. A duplicate-claim war in which eleven brokers sold the same finding under different names until buyers stopped believing anyone. And the false positive: a buyer who paid, on Vellum’s early advice, for a claim that did not work, then burned a source who had told him so.

“Automated volume is not a future risk,” Noa said. “It is last year’s ledger. That is why Vesper exists. Someone must price responsibility back into the market before the market drowns in green checkmarks.”

“And the flaw your people labelled derivative in June,” Sofía said. “The dependency parser. Five months unpatched

while your label priced it into a scheduling problem. My ex showed me the changelog. Who signed that label?"

Noa had known the question was coming since the changelog went public. She gave it the silence it had earned.

"It worked," she said finally. "The finding was real, Murmur produced it, and our classification buried it for five months because burying it protected the price of a human-origin package that was settling the same week. A person signed the label. I have read their reasoning and it was reasoning, not malice: the package's buyers were owed first position on anything in that family. The vendor patched late because of us. That one is on our ledger, and I will not dress it as anything else."

"Do you believe your own distinction?" Sofia asked. "Instrument, not author? Or is it commercially convenient?"

The question deserved honesty, and honesty had a floor. Noa looked at the provenance column still glowing on her screen, at the classification that had once, years ago in another life, attached itself to a young analyst's careful handwritten work because the analyst had made the mistake of being efficient in ways Vellum associated with machines. There had been an appeal. There had been a phone call Noa still thought about at odd hours. The analyst had left the industry, and Noa had stayed, and the difference between those two facts was the floor she did not show anyone.

"I believe it the way people believe things that cost them something," she said. "Which is the only belief worth selling."

She paid cash, both meals, visibly, and left the notes on the tray where the terminal could see them.

"One more thing," she said, standing. "Your five names. When you show them to the world, and you will, show the

error they share, not just the names. Names can be defended as coincidence. A shared mistake cannot. You learned that from the fork thread, so consider it a gift from someone who reads the same archives you do.”

She walked back across the square with her laptop bag over her shoulder, aware of the engineer watching her go, aware of a man at the kiosk who had been reading the same page of a newspaper for forty minutes, and no longer certain, which was new, whether the watchers around Sofía Alvarez were all working for people she could name.

Chapter 13: Murmur

The second message came through the same dead forum thread at noon: *Almagro, six o'clock. Bring questions you actually want answered.*

Mateo arrived early because the workshop taught him things his own rooms never did. At Lattice's rented compute everything was provisioned by ticket and nobody had ever held a soldering iron to anything. Here, a man was reflowing a laptop board under a magnifying lamp while arguing about football with the stereo woman, and the argument mattered exactly as much as the soldering. He had built a career on insisting that this kind of maintenance was nostalgia. He kept coming back anyway, which was data he declined to collect.

Sofía arrived at six precisely and did not smile at him, which was fair.

"I'll show you the harness," he said. "Not the production queue. The safe twin, the one we show auditors. You'll be able to see everything except the targets."

"That's a generous sentence," she said, "given what I saw your five names sign."

"They reproduced it," Mateo said, and heard how ready the sentence was, how it left him before thought. "Every claim runs green before it ships. Judge the result."

“You said that already.”

“It stays true.”

He set up the harness view on his laptop at the back table, angled as always, and let her watch Murmur work against a fictional target class. He loved this part, and he watched her instead of the screen while it ran, because her face did what everyone’s face did: resistance first, then unwilling respect.

Mateo still did not entirely believe the speed. That was the private fact he never offered in rooms where speed was his authority.

In 2022 he had photographed the first useful completion an editor gave him and sent it to Sofia like a magic trick. A line became a function. Then chat windows wrote files he copied by hand, reading the joins because the joins were where everything broke. Then the assistant entered the editor, read several files and offered a diff. Then it entered the terminal and ran the tests itself. Each change felt incremental while it happened. Looking back, the human role had moved from typing, to joining, to approving and to describing what success should look like.

Now he wrote a specification before bed and woke to candidates that had edited their own harness, failed, retried and ranked themselves while he slept. The first month, he checked the compute invoices against the dashboard because the number of attempted paths looked like a display bug. The second month, he checked less. By the fourth, disbelief had become an operational disadvantage. The people who paused to absorb what had changed lost a week to the people who treated astonishment as latency.

That was how intelligence got caught by acceleration. Not because clever people could not understand the new thing,

but because understanding took the same human hours it always had and the thing changed again before those hours were available. Mateo's doctrine made a virtue of the only response that kept him in the market: measure outputs, improve evaluations, stop demanding comprehension at the speed of production.

He believed the doctrine. He also needed it.

The loop laid out candidate attack surfaces, ranked them, generated test cases, executed, discarded, retrieved. Budget burned down the side of the screen like fuel. In forty minutes of wall time, compressed into eleven on the display, it covered ground no human team would survey in a quarter. And then it stopped on a finding, and even Sofia leaned in.

"There," Mateo said, unable to keep it out of his voice. "That path. Nobody put that in it. No template, no seed paper. It noticed that two components trusted each other's expiry checks and neither one owned the clock between them. It found a disagreement, the exact kind of thing you find, except it can walk ten thousand disagreements a night and remember every one."

"It's good," she admitted. Her voice did the thing it did when she was being precise rather than kind. "Now show me the queue behind it. Not the twin."

He had prepared for this. He brought up the honest version of the number: findings awaiting human review, plotted against review capacity. Two curves, one going up forever and one flat.

"The reviewers can't keep up," she said.

"No system's reviewers can keep up. That's not failure, that's scale. We sample. We prioritize by expected value."

"And the ones you don't prioritize?"

“Wait,” he said. “With status. Honest status, visible to buyers. Nothing is hidden, nothing is mislabelled.”

Except, said the part of his mind he kept fed and busy, that closely related variants of one finding had gone out under three separate names, priced separately, because no single human being at Lattice had ever seen the whole graph of what the queue contained, and he had known, and the knowing had felt like standing in current. Not hidden, nothing mislabelled. Reorganized. Optimized. He had told himself each sale was true in isolation, which was technically correct and therefore, by his own doctrine, irrelevant to notice.

On the bench behind them the stereo woman laughed at something the soldering man said, and Mateo thought of Micaela Ferreyra, whose evaluation notes on the dispatch tool were sitting in Lattice’s ingestion pipeline right now, flagged competent, queued to have a name attached to them at the moment of maximum value. She would say yes if asked. That was the thing nobody understood who called it theft: she would be asked, eventually, and she would be proud, and the pride would be the price.

“What happens to cleanup?” Sofía asked.

“Cleanup?”

“In the reproduction harness. When a run finishes. Something has to tear the environment down, rotate the keys it saw, delete the scratch state. Who wrote that?”

He knew where the answer lived. It lived in a module the loop had extended itself, eight months ago, past its original design, reviewed by an operator who checked the diff against the tests the same module generated, and it worked, it ran green every single night, and he could describe its contract in three fluent sentences and had not read it since spring.

"The harness owns its lifecycle," he said. "It's validated end to end. Every night it runs green."

"That's not what I asked." She waited, patient as a clock. "What does it do? If I asked you to explain, right now, line by line, what the cleanup code does with the material it holds, could you tell me?"

There was a silence of maybe four seconds, and Mateo heard himself cross it.

"You're asking me to do a machine's job," he said, warm and easy, buying the seconds with charm because charm had always been his working capital. "The whole point of the stack is that I manage evaluations and budgets, not lines. You don't ask a pilot to explain metallurgy."

"A pilot can tell you what the flaps do," Sofia said. "You just couldn't tell me what your own system does with a sealed target's leftovers."

And there it was, out in the open air between them, and he felt the strange relief of a deflection failing, like a tooth finally admitting it was bad. He did not answer. She let him not answer, which was worse than pressing, because she had learned that from watching systems, from knowing that a component that cannot explain itself will eventually fail in front of witnesses without any help from her.

She looked at his screen instead, at the burn-down curve resetting as the night's budget topped up, and he watched her watch it, and saw her recognize something before he did.

"Loss," she said quietly, pointing at the day's spent cycles. "Partial recovery, when a retry lands. An unexpected win, when the elegant finding surfaces. Increased stake." The curve climbed, reset, climbed. "One more run, Mateo. That's our trading screen. That's 2023. You're staring at the same

dopamine with better typography.”

“It isn’t,” he said, too fast. “Trading paid us for noise. Evaluations pay us with information. Every cycle produces signal whether it wins or not.”

“Some nights,” she conceded, and stood, and he understood that the concession was the whole argument: she had given him true things on both sides, and left him holding the difference.

At the door he stopped her, because some debts should be acknowledged in person.

“The five names,” he said. “When you use them. Remember they said yes. All five, real people, real signatures. Whatever you think of the machine that drafted their confirmations, the humans agreed to stand behind it. Nobody forged anybody. That has to count somewhere in your ledger.”

“Does it count in yours?” she asked. “That you needed five?”

Then she was out the door into Almagro’s evening, and he watched through the window as she walked three steps and stopped, the sudden full stop of a phone showing you something impossible.

He knew before he reached her, from the angle of her shoulders, roughly what it would be. A pre-authorization. A hotel in Neuquén, dated months ago, in her name, refunded flight attached, a journey she never took assembling itself into the commercial record one legitimate document at a time.

He had approved the dormant identity’s reuse in principle, in a document with a budget line, and had told himself that provenance was a market question and identity was inventory, and had not asked to see the individual plays. Standing three metres from her back, watching her read the record of herself

in a place she had never been, Mateo could not locate the exact moment when the doctrine had stopped being a door and started being a hallway with no other exits.

Chapter 14: Rosario

Declaration

The bus to Rosario took four hours, which was exactly long enough to draft a confession six times and delete it six times.

She had told herself she was going for ordinary reasons. Tuesday was her rotated day off; the incident was closed; families existed and required maintenance like any other system. But the bus window had the Paraná crawling past in the late light and she knew the truth of it: after the hotel in Neuquén she could no longer pretend her history was private property. It was contested ground now. You do not leave contested ground unvisited by the people who might have to defend it later.

Tomas met her at the door of his house in Fisherton with tongs already in hand. Her brother managed fire the way she managed incidents: calmly, with checklists nobody else could see. The asado had been planned for weeks, some cousin's birthday, and the patio held his wife's family, two children chasing a dog, and their mother holding court over the ensalada with the authority of a woman who had managed pharmacy stock through three collapses of the currency and considered all current crises poorly organized.

Sofía kissed her mother, admired the salad, played five minutes of football with a six-year-old and conceded four goals with theatrical grief. All of it was true and all of it was delay, and Tomas clocked the delay, because he had shared a bedroom with her for seventeen years.

It was nearly eleven when the last guests left and he began closing down the grill for the night, raking embers into the wind-down zone, closing the vents. She stood beside him in the smoke smell with a glass of wine she was not drinking.

"I need to tell you something about the account," she said.

"Banca Rocca." He did not look up from the coals. "What about it."

"You knew the name."

"I know every name you don't say twice." He moved a log end with the tongs. "Talk to the fire. I'm listening."

So she talked to the fire. It was not one confession but several small ones, and the fire kept interrupting, because a grill does not pause for narrative: he had to close a vent mid-sentence, move meat that was still cooling on the rack, test an ember with the back of his knuckle. She got the whole thing out in pieces between his maintenance. That she had never declared the account. That some gains from those years stayed undeclared too. That the regularizing she had described to him at Christmas, the clean paperwork he had toasted, was a plan she had not yet begun and possibly could not begin without costs she had spent four years not pricing.

Tomas closed the last vent and stood looking at the dead coals.

"How much are we talking about," he said, "in terms of my sister going to prison?"

"It wouldn't be prison. Probably fines. A settlement, if I

come forward first, before anyone hands them a file."

"Probably." He laughed once, without humour. "You know what I do all day? I tell people the exact percentage chances of things they're terrified of. I can't get one for my own sister because she's been managing me for four years with a story about a clean account."

"The account is real. The money is real. It came from my work, Tomas, not from anything else. I just never brought it fully inside."

"I don't care about the tax office's feelings." And here his voice changed, and she understood that the money had never been the wound. "I care that I reassured Mamá. Every time she worried, every asado where someone mentioned someone else's trouble with AFIP, I said *no, Sofía's fine, she fixed all that*. Do you know what that's worth? I lent you my credibility. You've been spending it for years without telling me there was nothing backing it."

The sentence landed harder than any number would have. Because it was true. She had let him believe the regularization had happened precisely so that his reassurance would stand between her mother and any question. She had converted her brother's trust into a buffer asset, the same way she converted everything, and buffers exist to absorb shocks that other people were meant to receive.

"I'm sorry," she said, and it was the first sentence all night she had not drafted on the bus.

"Is this the AI thing?" Tomas asked.

She almost said no. The account predated Murmur and Velum, and her tax exposure would remain if every model in the world went dark. But that was the kind of narrow answer she had come here to stop giving.

"It's part of why the old account matters now. Systems can produce more work than the people around them can understand. The market pays other systems to say the work was checked. My name is attached to some of those checks."

"Checks you did?"

"One I did. Others I didn't. The one I did was narrower than the way people used it."

Tomas put the tongs down. "You think medicine doesn't have this argument? The new anaesthesia station adjusts three things before I can reach one dial. It is better at holding a patient inside a range than the old machines were. I use it. I would be negligent not to. But when it alarms, the surgeon doesn't ask the station what it believes. Everyone looks at me."

"Because you understand it."

"No. Because I'm there." He wiped ash from his thumb. "I understand enough to know when to stop it and enough to answer for not stopping it. That isn't the same as understanding everything inside. Nobody does now."

The distinction bothered her because it was stronger than the position she had brought to the patio. Named responsibility did not require pretending one person contained the whole machine. It required locating the person who could interrupt it and the institution that gave them time to learn when interruption mattered.

"What happens," she asked, "when the machine changes faster than you can learn the alarm?"

"Then the hospital owes me training or it owns the mistake." Tomas picked the tongs up again. "And if it gives me neither, I don't sign a form saying the patient was safe because the graph looked smooth. Smart people get frightened too, Sofi. We just use longer words while it's happening."

He nodded, accepting it the way he accepted graft results, as data to work with rather than absolution. “First thing tomorrow I’ll send you the name of the accountant who saved Martín Delius. You call him this week. Not after Patagonia or whatever you’re walking into. This week.”

They went inside. Their mother was asleep in front of the television, and Tomas covered her with a blanket while Sofia watched, and nobody woke anyone.

She slept in her old room under the sloped ceiling, under shelves of books that had survived her leaving twice, and lay awake doing the arithmetic of who else had been spending her credibility. The memory came the way it always did now, uninvited and precise, wearing 2023.

Mateo at the kitchen table with both notebooks open, worried about the single route. Everything they earned abroad flowed through one processor, one bank, one wallet chain. If any node froze, they froze. Dependency on one path was a failure mode; she taught reliability, she of all people should see it.

And she had seen it, and her fix had not been diversification. It had been separation. Declared income through the front door, visible, taxed, boring. Verification proceeds somewhere else entirely, in the second wallet, attached to a second jurisdiction, with a paper wall between the two lives. She remembered proposing it across the pizza box, drawing the two columns on a napkin, *declared* and *actual*, and Mateo hesitating, actually hesitating, saying *are you sure you want to build it that way*, and her saying yes, because separation was the pattern she trusted: blast radius, containment, the failure that cannot spread. She had designed concealment the way she designed systems, well, and Mateo had gone along, and for three years

she had filed the whole architecture under *his idea*.

The napkin logic had even been correct, by its own lights. Contained failures protect the system around them. Nobody had asked whom the containment was protecting the system from being honest with.

In the morning Tomas made mate before dawn, standing at the counter in his hospital scrubs, and handed her the accountant's name written on the back of a shopping list.

"This week," he said.

"This week," she agreed, and it was even true, and they both knew the week had plans of its own, because her phone had buzzed twice against the nightstand overnight and she had not looked.

But that was tomorrow's problem. Today she hugged her mother, who patted her arm and said, with the terrifying indirectness of mothers, *you look thin, whatever it is, eat*, and Sofía said yes, Mamá, and meant it as a promise about more than food.

Chapter 15: The Other Sofía

The dossier arrived at noon by email, twelve pages, professionally formatted, from a company called Veritas Custodia that Sofía had never heard of and that had, apparently, heard a great deal about her.

She had sent them one line that morning, after a sleepless hour with the Neuquén pre-authorization: *I am given to understand your presence system places me at an event in Neuquén in April. I was not there. Please send me everything you hold.* It was the same instinct as auditing herself first: read the accusation before it reads you.

Veritas Custodia turned out to be Alero's presence-verification vendor, and their response was courteous, thorough and devastating.

April the nineteenth, according to their file, Sofía Alvarez had attended a sealed verification run at a facility outside Neuquén as an accredited human witness. Their evidence assembled itself across the page like a case study in how weak signals become strong proof. A flight to Neuquén, booked in her name, flown or not flown, refunded two days later in the way frequent travellers rebook. A hotel pre-authorization,

matching the refund window. A corporate expense approval from Traceframe's travel desk for *client verification support, Patagonia region*. A card token registered to her wallet, used at a fuel station on Route 22 at 21:34. And beneath it all, the anchor: a signed witness attestation, biometric badge log and a device attestation from a laptop whose hostname differed from hers by one character.

Every element was real. Each one came from a different commercial party, none of which had conspired, all of which had simply recorded what their systems saw. The journey had been manufactured out of true parts, the way a forged painting is made of authentic canvas, authentic pigments and one lie.

She sat in Tomas's kitchen with the printouts in rows and did what she did. She looked for the disagreement between records.

And found it, but not where she wanted. The fuel-station authorization was timestamped 21:34 local. The badge log put her inside the facility at 20:50 and 23:12. Route 22 ran forty minutes from the gate. The times were tight, impossible by car in both directions within the windows, unless the fuel stop happened before or after, and the receipt's own metadata, when she pulled the card statement from the frozen personal account, showed the token had been presented remotely, no terminal distance recorded, a contactless payment from a phone. Her phone, said the statement, because the token was hers. Except her phone had spent that evening, according to its own photographs, at a rooftop asado in Palermo, camera roll timestamped against network logs.

One lie in a painting of truths. But when she marked up each element as real or fabricated, the pattern refused to settle into a single author. The flight and expense approval smelled

like a travel platform compromise, which meant Lattice. The card token and remote payment smelled like compensation-channel work, which meant Vesper's intermediary. Two forgers, building one woman, each supplying half the canvas, neither apparently aware of the other's brushstrokes. The other Sofía was a joint production.

Then she reached page eleven, and stopped being able to breathe evenly.

The witness attestations were not all from April. Veritas held seven runs across fourteen months. On four of them, the signature block carried a witness credential issued in 2023 to Sofía Alvarez. Not a forgery of her signature. Her actual credential, her actual key, signing actual sealed-test attestations at actual facilities, months apart, in a hand her hands had used exactly once, at midnight, three years younger, for a bonus expiring at dawn.

She had assumed the dormant identity was stolen. Stolen she could fight; stolen was someone else wearing her. Page eleven said worse. The key had been preserved, maintained, kept alive like a heart on ice, and then used, really used, on real runs, by whoever held it now, so that some fraction of the recorded Sofía was genuine work performed under her name. She would never be able to prove which attestations were the fabricated woman and which were her leaked self doing competent labour in a sealed room in Neuquén. Neither, she realized with a cold that started in the spine, could Alero. Neither could the market. There was no longer any archive that could separate the two women, because the two women shared one signature.

The phone rang at three. San Marino country code.

"Señora Alvarez." The voice was low, precise, accented to-

wards Italian Spanish. "Rina Belluzzi, Banca Rocca client services. We have spoken twice in four years. I am calling against procedure, from a mobile, during my break."

"What's wrong?"

"Three things, briefly, because my break is fifteen minutes." Papers shuffled. "First. A replacement card for your account was requested nine days ago, through the app, passed identity review, delivered to an address in this republic that is not yours, and activated. You did not request it?"

"You know I didn't. My cards are frozen. I froze everything the Friday my lunch bought itself."

"Correct. The request passed anyway, because it answered every challenge correctly. Device, history, behavioural questions. Whoever requested it knows your patterns at our bank better than I do, and I have read your file." A pause. "Second. A vetted intermediary has been querying files connected to your account. Vetted means contracted through channels I am not permitted to describe, for purposes nobody will write down. My institution answers these queries. It is not wrongdoing on our side. It is the pipeline working as designed, which is worse."

"And third?"

"Third is me telling you this, which risks my employment, which is why the call is short." Rina's voice flattened into professionalism. "I am not your ally. I am not anyone's ally. I am an officer of a bank with a client file that has begun to generate irregular queries, and there are procedures for what I do next, and I intend to follow them, because procedures are the only thing in this story that cannot be threatened."

"The fraud record," Sofia said. "The card request, the queries. Preserve it. File it formally, date-stamped, inside

your own system, even if nobody asked you to. Especially if nobody asked you to."

"That creates a document I may have to justify."

"Yes. And if it's ever needed, it exists, and you can prove when you made it. That's the whole point of you, Rina. Be the clock."

The silence on the line lasted long enough that Sofía heard the break room around her: cups, distant Italian, a television.

"My shift ends at five," Rina said. "Preservation requests filed before close are processed same day. Good afternoon, señora."

Afterward Sofía sat at Tomas's table among the twelve pages and understood the shape of the trap she was standing in. Because the false Sofía was, by now, easier to believe than the real one. The false Sofía had no second wallet, no midnight signature on a job whose cleanup she had narrowed into defensibility, no undeclared account generating irregular queries. The false Sofía travelled, witnessed, signed and paid for fuel with clean tokens. If a tribunal someday laid both histories side by side, the fabricated woman would look like the better person, and the only evidence the real one could offer in her own defence was the confession she had already given her brother in front of a dead grill.

Outside, Tomas's neighbour washed a car in the afternoon heat, radio on, some cumbia hit from another summer. Ordinary world, going about ordinary hours. Sofía gathered the pages, squared them, and put them in order by timestamp, because that was the one act available to her that no forger could touch: decide, before anything else happened, what order things happened in.

Chapter 16: The Clock

The two requests arrived within four hours of each other on Wednesday, and each was individually the most boring email of Sofia's week.

She saw them together only because Tomas's guest-room wifi dropped during an afternoon storm and her phone fell back to mobile data, where the audit federation's ticket queue loaded in plain text, oldest first. The federation was the quiet achievement of Martin's best years: seven write-mostly audit sinks scattered across the hemisphere, each stamping event streams with time it owned, each administered by an engineer who had keys to a building and a duty to be annoying about procedure. Sofia was the designated operator for the Paraná node, a half rack in a cold aisle of a provincial colocation facility that had survived three owners and every flood since 2012.

São Paulo had convicted the first impossible receipt. Paraná was the regional collector: it retained São Paulo's signed stamps alongside the later witness streams, enough history to show whether nine minutes early was an accident or a practice.

Ticket one, filed 14:12 by a subcontractor she had never heard of, on behalf of a client she was not permitted to see: an exported reset of the node, citing infrastructure migration.

Full state transfer, then wipe and rekey. Paperwork immaculate. Signed by two people whose titles were longer than their names.

Ticket two, filed 17:48 by a facilities contractor booking emergency maintenance on the node's rack, power circuit and network uplinks, Thursday 06:00. Also immaculate. Cited a thermal incident in the neighbouring aisle.

Alone, either request would have sailed through on Friday morning while she slept in. Together they were a pincer. The reset would hand the node's entire honest chronology to whoever received the export, and the maintenance window would put new hands on the machine before anyone could ask what the first request had taken. She did not need to know which faction stood behind which ticket. The choreography told her enough: both sides wanted the clock, and neither knew the other had booked it.

She checked the queue's approval state. Both tickets sat at stage two of four. Automatic checks passed, human sign-off pending, scheduled execution Thursday dawn. She had one night.

She booked the last one-way rental available from Rosario and called Tomas.

Tomas drove her to pick up the rental car at nine and asked no questions, which cost him visibly. At midnight she crossed the river north out of Rosario, the bridge deck vibrating faintly under the wind off the Paraná, the water below running wide and black and indifferent as an unlogged stream. Paraná itself lay two hours further, most of it on empty dual carriageway through dark farmland, trucks for company, her coffee going cold in the cupholder.

She drove badly by her own standards, twice catching her-

self doing a hundred and thirty with her thoughts somewhere else entirely, and made herself slow to the limit, because arriving mattered more than arriving fast and because a speeding fine in her name tonight would fit somebody's file a little too well.

The colocation facility hid behind a logistics park at the edge of the city: grey cladding, a single lit window where a night guard sat with a television murmuring, a roller door, a mantrap, cameras that had never been networked to anything interesting. Her keycard opened the outer door because her keycard had always opened the outer door. That was the whole of her superpower, and she leaned on it now. She was not a spy. She was the administrator. The building's own records would show, from tonight forward, exactly who had entered, when, and why, and she filled in the visitor log with her real name and the words *quarantine action*, *ticket review*, because the facility deserved true entries even if nobody else's week contained any.

The node lived in rack B4, between a grain exporter's telemetry stack and a municipal fibre switch. Four machines, no monitors, lights steady as sleeping animals. She crouched in the cold aisle with the maintenance manual open on her phone and worked by procedure, the way her hands did everything that mattered.

First, stop the intake feeds, cleanly, with notifications to the federation list, so the stopping itself became part of the record. Second, unmount the write path and remount the volumes read-only, so that nothing, including a mistake of hers, could alter what was already stamped. Third, pull the network cables and coil them on top of the rack in plain sight, an act so old-fashioned it circled back to sophisticated: no port scan de-

feats a cable lying on a shelf. Fourth, image both storage drives to fresh disks from the spare kit she carried, hashing each image twice, labels written by hand.

Two hours, start to finish. The facility hummed around her. Once, near the end, headlights swept the loading bay, slowed, and moved on, some truck choosing another door, and she knelt motionless between racks with a screwdriver in her hand and her heart going like the failed drive in Julián's pool, and after a minute the diesel smell faded and the hum resumed.

At half past three she stood, cracked her back, and looked at what she had made. A sealed archive, deaf and blind, holding every honest timestamp the region had fed it since spring. Two images in padded cases, one destined for a safe deposit in Rosario come morning, one for a drawer in Palermo, neither for her apartment, neither for Traceframe, neither for anywhere with her name on the lease. Honest copies work only when custody is boring.

She wrote the quarantine entry in the facility log, signed it, photographed every page, and emailed the federation from the parking lot at 03:51:

Paraná node quarantined at operator discretion pending review of tickets F-2211 and F-2214. Intake halted, volumes read-only, uplinks physically disconnected. Images taken. Execution of either ticket will require my physical presence and my keycard. I can be reached, but not quickly.

She watched the mail leave, then turned the phone off entirely, first time in six years, and sat in the rental car in the empty lot eating biscuits from a petrol station, and let herself feel the size of what she had done.

Because here was the joke the week had built. Both factions held dossiers placing her in Neuquén, Recoleta, two cities at

once, a woman woven from real records into impossible journeys. And the one journey she had actually made, tonight, alone, documented by a guard's logbook, a visitor signature and a camera nobody had thought to suborn, was the one record in the entire war that was beyond forgery, because it had never been interesting until now.

They had spent months manufacturing her presence everywhere she had not been. She had just spent two hours establishing it where she was.

The sun came up over the logistics park like a slow boot sequence. She drove back towards the river with the two cases on the passenger seat belted in, thinking about the sentence she had just signed into existence with her own hand: *execution will require my physical presence*. By nightfall, both sides' ticket systems would have parsed it. Neither faction could touch the clock without collecting Sofía Alvarez, in person, at a door she controlled.

For the first time since the first impossible receipt, being the prize felt like having leverage. She took the first exit for fuel, bought two more coffees and a bag of facturas, and pointed the car towards the sunrise and her brother's kitchen table, where a telephone was almost certainly already ringing.

Chapter 17: The Israeli Story

The story reached Sofía the way most stories reached her that week: as a document she had not asked for, attached to an email she did not trust, from an address that had never written to her before.

The sender called itself Observatorio Sur, which sounded like a think tank and was, on inspection, a domain registered eleven days earlier. The attachment was a briefing titled *Alero, Vesper and the Southern Corridor: Notes Towards a Public Record*. Forty pages. Charts. Photographs of Noa at three different conferences, each captioned with the name of an institution she had once worked near. A timeline that began with bilateral agreements signed in Buenos Aires and ended, without any visible join, at a sealed facility in Neuquén province where, the document said, Israeli-origin verification technology was being battle-tested against regional targets under the cover of commercial research.

It was beautifully made. That was the first thing she noticed, and the first thing that frightened her, because she had spent the week reading fabricated records and this one had the same quality: every individual element real, the joinery invisible.

The conferences were real. The bilateral programme was real; she could name two colleagues who had attended its workshops. Noa's history was real. The photographs were real. The conclusion, which sat on page two where a busy reader would meet it first, was not supported by anything else in the file, but it no longer needed to be. By page two the reader had already stopped checking.

She did what she now did automatically. She read the accusation last and the citations first, and she traced each citation to whatever custody it claimed.

The bilateral events checked out and led nowhere near Piedra Sola. The programme's published partner list contained no Alero, no Veritas Custodia, no witness vendor of any kind. Investor travel records, insofar as the document summarized them, showed ordinary deal-flow between Tel Aviv, Madrid and Buenos Aires, the same circuit every venture newsletter covered openly. The one claim that pointed at the facility itself cited a source described only as *documentation reviewed by our researchers*, which was the citation equivalent of a receipt whose environment did not yet exist. She wrote *no supporting link* in her notebook, next to a list of everything the story needed to be true, and the list was entirely composed of things that were merely plausible.

Mateo called at six, from a road, wind past his window.

"You've seen the corridor piece," he said. Not a question.

"I've seen a document."

"Everyone's seen it by tomorrow morning, that's how these work now. One well-made file, seeded in four places, and by the weekend it has citations of itself." He let the wind fill the line for a moment. "I want you to know it came from our side of the fence. And I want you to sit with why we're the ones

who pushed it.”

“Because you want it to be true,” Sofía said.

“Because it would work.” No apology in it. “Vesper is a company built inside a political relationship. Their access, their contracts, their whole moral posture about custody, it all breathes through those ties. You can’t prove witness fraud against them in a technical venue; their reviewers are too good and too patient. But a political scandal doesn’t need proof, it needs plausibility. It freezes them for a year. It freezes their premium, their procurement, everything. One document does what none of Murmur’s findings can do.” He paused, and when he continued his voice had dropped its salesman register. “You’ve read it. Tell me it isn’t tight.”

“The events are real. The link doesn’t exist. There’s nothing between the photographs and the facility except the layout, and the layout was designed so you’d stop looking.” She heard him exhale. “You taught me spreads, Mateo. This is a spread. Every leg is real and the arbitrage is fiction. You know exactly what you’re selling.”

He was quiet long enough that she thought the call had dropped.

“He says no,” Mateo said finally, to someone in the car who was not there. “All right. All right.” And then, to her: “For the record, I didn’t write it. I just hoped. There’s a difference, and I’m aware how thin it is from where you’re standing.”

“It’s the thinnest part,” she said, and hung up, because he had given her something worth more than the argument: confirmation of whose narrative the document served, volunteered by the man who wanted it.

Julián’s email arrived at half past nine that night. Two lines of text, a spreadsheet, and no greeting beyond her name, be-

cause he had promised one comparison and he delivered exactly what he promised and nothing more.

Did the thing you asked. The mirror kept everything. Dates below are commit times from our side, which nobody outside the room could have adjusted after the fact, because nobody outside the room has ever had the key.

She opened the spreadsheet and found the chronology of a single zero-day claim: when it first appeared in discussion, which fork branch held it, when it moved, and when Alero's public-facing receipts, pulled from the trial notices Martin had once followed, dated its certification.

The claim had lived in the branch that accepted disclosed assistance with human sign-off. Then it moved to the branch that banned AI-origin reports entirely, and its authorship trail vanished in the crossing, exactly as they had reconstructed behind Cable Rojo's rehearsal room. But the dates told the rest. The move between branches preceded Alero's earliest receipt for the same claim by nineteen days. The discovery had been finished, discussed, forked and re-labelled before either faction's machinery had ever certified that it existed. The receipt had not witnessed the finding. The receipt had arrived to collect it, the way a delivery van arrives after the furniture is already on the pavement.

She sat with the spreadsheet open for a long time, feeling the shape of the whole thing settle into place around this one clean column of dates. It was the first artefact she owned that proved pre-minting from outside the pods themselves. Julián had given her exactly one comparison and it was enough.

At the bottom of the mail, below the signature-free sign-off, one further line, added apparently against his own policy of saying less:

Whatever this is, it looks worse than your ticket queue. Don't sign anything that feels satisfying.

She drafted a reply, deleted it, and sent him instead: *Received. Thank you for keeping the archive.* He had asked nothing about her finances or her love life, and she was not going to spend his goodwill on either.

The corridor document was still in her inbox, forty pages long, and she caught herself reading its conclusion again, feeling the pull of it. Because the story was useful. If she believed it, everything simplified: one villain with a flag, one scandal, one freeze. It was the most attractive inference anyone had handed her since this started, and it required her to sign precisely the thing the evidence did not support, which made it a rehearsal. Somewhere ahead of her was a room where people with badges would ask her to state what she knew, and the question was whether she would say what was true or what worked.

She filed the document into the folder where she kept things she refused to believe yet, next to Rina's warnings and the twelve Veritas pages. Then she put Julián's spreadsheet beside it: two lines of introduction, dates held by a mirror nobody had thought important enough to buy.

One account needed forty pages to cross the distance between a conference and a crime. The other needed a column of times. On Monday night, she knew which one she was willing to carry into the room.

Chapter 18: No Quorum

With the corridor document refused and filed, Sofía finally built the overlay she had been carrying around in pieces for a week.

She did it on the kitchen table in Palermo, the one surface in the apartment that wasn't also a screen. She taped together four printouts, opened her notebook to a blank spread, and drew the timeline by hand before letting any software near it. Four layers, four colours.

Only after the sources were fixed did she give a redacted table of times to Traceframe's approved analysis assistant. She asked for interval overlaps, clock-domain conversions and duplicate identifiers. No authorship. No motive. No instruction to decide what the pattern meant.

It returned eleven candidate crossings in thirty seconds. Eight vanished under review: daylight-saving assumptions in foreign records, one duplicated receipt, two event times confused with processing times. Three survived. Work that would have taken her most of a night had taken an hour, including the time required to disprove the assistant's confident mistakes.

At the bottom it added a finding she had not requested: *Pattern is consistent with coordinated state-level pre-positioning, con-*

fidence 0.81. Its citations were the bilateral programme, Noa's travel and the Observatorio Sur document, three repetitions of the same unsupported join. The assistant had crossed the distance from useful arithmetic to geopolitical accusation without experiencing distance at all.

She deleted the conclusion and kept the three intervals. Bounded assistance did not mean the tool stayed inside the boundary. It meant the person using it knew which outputs had crossed.

The first layer was the receipt chronology: every Alero certification she could legitimately see from the incident and the federation's own notices, including the nine-minute case that had started everything.

The second layer was Lattice's authorship record, reconstructed from chapter and verse of what Mateo had shown her and what she had traced herself: findings generated by Murmur, human names attached afterwards, variants of one discovery sold under separate labels because no single reviewer ever saw the whole graph.

The third layer was Vellum's: Vesper's adversarial pipeline fingerprinting model output, generating hostile cases, feeding dead ends into Lattice's queue, and labelling genuine rival discoveries as derivative when provenance mattered more than effect.

The fourth layer she drew last, in red, because it belonged to neither faction. It was Martin's mixed record, the thing he had built and tried to sell twice, whose existence she knew from his trail and whose contents she still had not seen.

She laid the layers over each other and looked for the disagreement between records, the way she always had, and this time the disagreement was not a discrepancy hiding in one

system. It was the whole architecture.

Because the two witness populations were supposed to be independent. That was Alero's entire product. Quorum meant that separate parties, with separate incentives, operating separate appliances, had each certified the same sealed test without collusion, and the market paid Alero precisely for that separation. Vesper supplied human-provenance witnesses. Lattice supplied model-side witnesses. Neither trusted the other, which was the guarantee.

Except both sides had spent the past year pre-minting inventory. Lattice minted receipts ahead of scheduled runs so variants could later sell as distinct discoveries with clean chronology. Vesper ran warm replacement appliances so its human-witness line could certify on schedule regardless of when anything actually happened. Each side assumed its own schedule was the private one, and each side built its forward-dated records against the other side's *announced* calendar.

Their schedules collided.

She found it in three places on the overlay where the colours crossed. At each crossing, a Vesper witness and a Lattice witness had certified the same test, both records dated correctly by their own clocks, both fabricated, and neither fabricator aware that the other had booked the same future. The quorums at those crossings existed. They were valid by Alero's rules. And they were controlled by nobody, because the two histories certified each other. A Vesper witness had put its signature under a Lattice fiction; a Lattice witness had witnessed a Vesper appliance running hot in a crate. If either faction exposed the practice, it exposed its own.

That was why neither side could stop the corruption now. Not secrecy. Mutual hostage.

Martin's red layer explained itself once she saw the crossings. He had found the same collision months ago, from inside, as a human witness who kept honest notes, and instead of reporting it he had built a third record: a continuity file mixing elements of both frauds, provable enough to hurt either side, held by him alone. Two buyers, mutual exposure, neutrality through opposing fear. His flaw, laid out on the kitchen table in coloured ink, was exactly as simple as she had suspected since Toronto went quiet.

At noon, the public confirmation arrived without asking her. Alero published a ceremony notice, ordinary in form and extraordinary in date: an emergency root ceremony, Monday night, at which all existing receipts would be migrated under new witness keys ahead of *settlement scheduled for Tuesday*. The largest AI-generated zero-day auction in the market's short history, settling through conventional bank legs and stablecoin escrow at once. Every old receipt, including every crossing, every pre-minted forgery, would join the new root. After Monday, the mixed history became permanent infrastructure. Refuse the migration and dangerous submissions returned to uncertain custody, which served nobody except whoever wanted the market dark.

Four days.

Her phone rang while she was photographing the overlay. Seattle area code.

"Sofía. It's Daniel." A pause she had learned to hear as him choosing words with both hands. "Two things. First, official. Traceframe has been invited to a customer review session at Piedra Sola next week, part of the standard pre-settlement assurance our contract entitles us to. I'm authorizing you to attend as our technical delegate. You'll have legitimate access,

legitimate cover, and a legitimate reason to be in the building.”

“And the second thing.”

“The second thing isn’t official.” Paper moved somewhere near his desk. “I’ve been reading your incident file again. The one I told you to restore service on. I want you to know that I’ve re-read the exception language, and I’ve asked counsel for a formal interpretation, and I’m not ready to say more than that before Monday.” Another pause. “Your name is already on Alero’s human-witness list, by the way. Has been since April. You may want to think about what you’re going to do when someone in that building asks you why.”

“I didn’t sign up in April.”

“I know,” Daniel said. “That’s what worries me about how well it will fit.”

He hung up before she could answer, and she sat looking at the phone, then at the overlay, where the red layer crossed both coloured frauds and belonged to a missing man.

She thought about Julián’s warning. *Don’t sign anything that feels satisfying.* Everything about the coming week would feel satisfying. Exposing Vellum would satisfy Mateo’s account of the world. Exposing Murmur would satisfy Noa’s. Accepting either satisfaction would wire her, signature first, into somebody’s settlement ledger.

What she could actually prove was smaller than any of it, and she wrote it out in her notebook in plain words, so she would recognize it later under pressure:

A quorum requires witnesses that coexist in time. During these intervals they did not. Therefore no valid quorum formed. Nothing more than that. Not what the exploits are worth, not who authored them, not what either company deserves. Only when.

It was not satisfying at all. She read it back twice and made

herself stop looking for a way to make it bigger.

She closed the notebook and left the overlay drying on the table. The red line crossed both factions at three points, holding the only complete account of the collision that anyone, on either side, had bothered to assemble.

Four days from now, every line on the page would be invited to call itself one history. She had until then to keep them separate.

Part Three: The Forty-Seven Minutes

Road, Aeroparque, Piedra Sola. Four days before settlement.

Chapter 19: Gate Thirty-One

Mateo crossed into Río Negro on the second day of driving, with three witness appliances in padded cases wedged behind the seats of a rented pickup and the air conditioning fighting a losing war against the Patagonian afternoon.

The hardware was real. That much he kept telling himself, and it kept being true. Sealed units, current firmware, custody chains that would survive an audit, each appliance capable of doing exactly the work its documentation described. He had checked every unit himself before loading them, because whatever else he had become this year, his kit still worked.

The collective was another matter.

He had promised Lattice's logistics desk a delivery to the *Colectivo de Investigación Abierta del Sur*, eleven researchers, stateless by choice, operating outside both factions' provenance regimes. The collective existed in the sense that a group chat existed. It had a name because names were free. Its membership was four people, two of whom had never answered a message since March, one of whom was primarily a photographer, and a student in Bahía Blanca who believed, with a sincerity Mateo found almost unbearable, that open

witnessing could outcompete both Vesper's human premium and Lattice's volume. Nobody in the collective could service a witness appliance. Nobody had asked for three.

Lattice wanted the delivery made and photographed. Excluded researchers receiving real hardware made good material for the account of the world where Lattice was infrastructure for everyone the incumbents had priced out. The photograph would be true. Everything around it would be arranged. He had built a career on insisting that arrangement was not lies, only selection, and the sentence had begun to taste like fuel.

At a fuel stop outside Sierra Grande he bought coffee and stood at the counter doing the arithmetic he had been avoiding since Buenos Aires. The appliances in his truck, if they reached their scheduled runs, would certify on dates already chosen for commercial convenience. He knew that now. He had known it since the collision surfaced, since the crossing quorums appeared in the reconciliation and nobody on his side moved to burn the inventory. His own doctrine said origin did not matter; output was true or false regardless of who produced it, and the market should judge the work, not the worker. But what he was driving north was not output. It was future agreement, sold before any test existed, and there was no version of *judge the work* that survived that. He had spent two days looking for one, because the alternative was to admit that Sofía had been right about him three years ago, in a rented flat in Córdoba, when she told him he fell in love with systems faster than he learned to maintain them.

He drove on. The road ran straight through country that did not care about his doctrine: wind farms turning slow against the plateau, a school bus, sheep fences running to

the horizon, a town whose water tower carried somebody's painted name. People lived here. The trucks came through here. The facility he was delivering towards sat inside all this ordinary life like any other tenant, paying rates and employing guards, which was somehow more damning than if it had floated in emptiness.

His phone, mounted on the dash, showed the delivery confirmation form. Signatory: collective representative. He had listed himself.

Sofía's ticket cancelled itself while she was in the security line.

She felt the phone buzz at 10:12, boarding pass already in her wallet app, gate Thirty-One at Aeroparque, the Neuquén flight, and the cancellation notice arrived with the cheerful finality of a system doing its job: *schedule change; new itinerary to follow*. No new itinerary followed. Her booking, made Tuesday through Traceframe's travel desk under Daniel's authorization, simply ceased.

She had budgeted for this. The second seat existed before the first died: a personal fare, different carrier, bought Wednesday from Rosario with her own card, the card Rina's institution knew about, which was the point. Two valid boarding records for the same woman, same morning, same airport, different routes. Whichever record the watchers chose to believe, the other one would contradict it, and she would know who held which file by how they reacted.

Because there were watchers. She had counted on that too.

Noa she expected and found: business class lounge guest, legitimate consulting cover, a small nod across the terminal

that committed neither of them to anything. The other two were less polite. Sofia ran her own route through the terminal the way she read clocks. She walked past the kiosks once without stopping, doubled back for coffee she did not want, changed direction at the last moment before the moving walkway, and watched the correlation.

Two men adjusted. Not together. One, grey jacket, tall, broke off his pretend phone call within a second of her reversal and drifted towards the gate boards. The other, shorter, denim, took longer, maybe four seconds, and over-corrected, ending up ahead of her instead of behind, which professionals did not do unless they had just inherited the tail from someone else's handover. Grey jacket moved when she moved, smoothly, bored-looking, rehearsed. Denim moved when grey jacket moved.

That interested her more than being followed at all. She tested it twice more at the gates area, small changes, a false step towards the toilets, a pause at a bookshop window, and the pattern held. Denim was not watching her. Denim was watching grey jacket, and correcting around him, and grey jacket either had not noticed or had noticed and considered denim someone else's problem. Two factions, each certain it owned the observation post, each the other's blind spot, and she was the overlap. Neither of them could report her movements without reporting the other one's presence, which meant both reports would arrive contaminated, and contaminated reports created hesitation, and hesitation was hours. She could spend hours.

She bought water and a sandwich and ate half of it at the gate, deliberately visible, deliberately boring, letting both men log a woman waiting for a flight that, according to one of her

two boarding records, she was not on.

Noa passed close enough to speak without stopping. "Grey jacket is yours," she said, in English, eyes forward. "Mine doesn't fly until tonight. Enjoy being popular."

"Which side is he?"

"No idea. Ours stopped pretending to be clever about it months ago." A flicker of something crossed her face that was nearly humour and mostly tiredness. "You know what happens Tuesday."

"I know what I can prove."

"That's not the same as knowing." And she was gone towards the lounges, exact and unhurried, a woman whose market was dying in four days and whose walk did not show it.

Boarding began at 11:40. Sofía used the personal ticket, the personal card still loose in her pocket where she could feel it, and watched the gate agent scan her through without interest. Behind her in the jet bridge, grey jacket boarded too, three places back, no longer pretending. She did not look for denim. Somewhere in the terminal he was composing a report about a man on a flight he was not booked on, and the report would take time she intended to use.

The plane pushed back into a wind that rocked it on the apron. Through the window the city slid away, the river flat and brown, and Sofía opened her notebook to the page with the plain sentence written in ink and read it once more, because on Monday night, in a sealed hall at the end of a long road, it was going to be the only thing she had that everyone else in the room wanted her to make bigger.

Chapter 20: The Alero Hall

Lucia met the Traceframe delegation's van at the inner gate because nobody else on the morning shift wanted the escort duty, and Lucia took escort duty the way she took everything: seriously, quietly and with her own checklist.

"Four visitors," she said, counting them against the manifest on her tablet as they climbed out into the wind. "Two engineers, one counsel, one delegate. Delegate Alvarez, you're with me first. The others go to briefing room two until your appointment."

The delegate was younger than the file photograph and moved like someone reading the ground for tripwires. Lucia had read the file too, all of it, including the pages Alero's own presence system could not explain. She kept that knowledge where she kept everything: in the shift plan, not on her face.

Piedra Sola spread below the ridge road as they walked: the energy campus first, turbines and switchgear and the maintenance sheds that fed her family, then the newer buildings, then, at the far end, the hall itself. Low, grey, deliberately boring, its roofline broken by heat exchangers rather than anything dramatic. The wind came off the plateau

in long steady pushes and the flags on the admin building cracked like whips.

"You're from Bariloche," Sofía said. It was on Lucia's badge, under the name.

"Born there. Educated in Neuquén. Came back because industrial systems have answers." Lucia walked half a pace ahead. "Everything here answers eventually. The trick is asking questions the equipment can survive."

Inside, the hall divided into what the brochure would have called modest compute and what Lucia thought of as the honest part. Rows of ordinary racks, mid-range accelerators by the pallet, nothing frontier, nothing that would make a headline. The instrumentation was the expensive part. Sensors on every circuit, power meters on every socket, environmental telemetry sampled faster than most labs sampled their actual experiments. Alero sold receipts, and receipts required proof of conditions, so the building measured itself the way a jeweller's scale is calibrated: constantly, jealously, against references nobody else was allowed to touch.

Along the far wall, sealed doors in pairs. Behind each pair, a digital twin environment, an isolated replica of some customer's system built to take real attacks without any real world attached. The twins were fictional to everyone outside their walls; even Lucia's access stopped at the doors.

"The targets are confidential," she said. "I don't know what's behind them and neither do you. What I know is power draw, thermal load and schedule. That's the whole of my visibility, and I'll thank everyone this week not to assume it's less."

"And the witnesses?" Sofía asked.

"Witness pods are the small cabinets in the witness corri-

dor. Sealed appliances, one per tenant faction plus independents, each with its own clock, its own printer, its own custody log. They watch the tests through instrumentation feeds only. No exploit contents cross their boundary. A receipt says *a test of class X ran under conditions Y between times Z*, signed, and the buyer trusts it because four independent machines say the same thing.” Lucia stopped at the corridor mouth and looked down its length: eight grey cabinets, status lights steady, printers silent. “That trust is why we’re all still employed.”

They walked the floor while Lucia did her rounds, and she talked the way she always talked to visitors who asked real questions: about the campus beyond Alero, the legitimate energy-modelling work running normally in the buildings they had passed, the wind farm’s maintenance calendar, the crew from Puerto Madryn rewarming a transformer, local objections to truck movements that she had personally taken to Esteban’s office. Patagonia worked here. The facility was a tenant inside a working region, and she was the person who made the region and the tenancy hold together.

What she did not put in the visitor version was the payroll meeting two winters ago. The materials lab had lost its largest customer and Esteban had placed two columns on the screen: jobs without Alero, jobs with it. Nine names survived in the second column. One belonged to Lucia’s night electrician. Another belonged to the woman who calibrated the hall’s airflow sensors and had just taken a mortgage in town. Tenant confidentiality, Esteban had said, was not an abstract virtue when it paid salaries on the last business day of the month.

Nor was Alero the only acceleration on campus. The youngest instrumentation technician used a model on his phone to translate manuals and propose diagnostic sequences

for equipment whose manufacturer no longer answered Argentina. He could repair boards Lucia had once shipped to Germany. The oldest machinist had stopped receiving drawing work for guards and brackets because customers generated acceptable replacements from photographs and sent them directly to automated mills in Buenos Aires. He now drove the campus shuttle three days a week and refused to call the change progress or theft. It was simply the fact that his hands were worth less to people who had never watched them work.

Lucia did not have one opinion that covered both men. She distrusted anybody who did. What she knew was consequence: new capability arrived in a week, new procedure in a quarter and new training after somebody found a budget. The gap was staffed by whoever happened to be standing closest when the alarm began.

If Alero fell, nine innocent names might fall with it. If it stayed under its present contracts, Lucia's signature would keep protecting a customer whose work she was no longer willing not to understand. That was the choice waiting under Monday's ceremony, though nobody had put it on the manifest.

Sofía listened properly, which Lucia noticed and filed away. Most Buenos Aires visitors heard *distance* and stopped listening, assuming emptiness followed. This one asked what the night shift cost, who supplied the diesel backup, how far the nearest hospital was and whether the road closed in September.

At the witness corridor, Sofía paused.

"Can I see the delivery logs? For the replacement appliances arriving before Tuesday?"

"You can see the summary page. Contents stay sealed until custody review." Lucia pulled the tablet up and turned it around. Two replacements scheduled, Saturday delivery, one from Vesper's supplier, one through a Lattice reseller, both already manifested, both already assigned to Monday night's ceremony.

Something crossed the visitor's face that Lucia recognized from her own mirror: the look of a person reading a document and finding one line missing from where it should be.

"Problem?"

"Your ceremony is scheduled for Monday midnight into Tuesday," Sofia said. "The auction settles Tuesday evening. Those are very close together for a root migration that size."

"They are," Lucia agreed. "I've asked twice why the window is so tight. Both times I was told settlement drove the date." She put the tablet away. "I don't like answers I can't check against a purchase order. But that's Tuesday's problem, and today I have a transformer to collect."

She left the delegate at briefing room two with the rest of the Traceframe party, nodded to the corridor guard, and went out into the wind to do the part of the job that had answers. Behind her the hall hummed, measuring itself, and eight witness cabinets stood in their corridor printing nothing yet, holding their clocks like held breath.

Chapter 21: Warm on Arrival

The replacement appliances arrived Saturday at 09:20, two hours before their manifested window, which was the first thing Lucia wrote down.

She met the courier truck at the goods bay with the custody log open and made the driver wait while she photographed the seal positions, the case serials and the truck's temperature dashboard, in that order. The driver, a contractor from Neuquén who had delivered here before, waited with the patience of a man being paid by the hour.

"Two units," he said. "Signed manifest. One for your Vesper channel, one through a reseller, don't ask me whose."

"I'm not asking," Lucia said. "I'm weighing."

Because that was the second thing: the forklift's load reading. She had calibrated it herself on Thursday. A sealed witness appliance with its battery conditioner weighed eleven kilograms and four hundred grams, plus or minus the mass of the desiccant pack, and she had the certificate on her clipboard to prove it. The left-hand case came off the truck heavy. Not much. Two hundred grams, maybe three. The right-hand case came off within spec.

She carried both into the intake room alone, per procedure, and began the examination she had performed once before in this building, three weeks ago, on a delivery nobody had thought important.

Filters first. A sealed appliance that has never run carries clean filters. The left-hand unit's intake filter held dust. Not much of it, but Patagonian dust has a signature, fine volcanic grit from the plateau roads, and there it was under her loupe, grey-pink, dry, exactly what three days in a warehouse near the highway would deposit. The unit had breathed filtered air somewhere that was not a factory floor.

Battery cycles next. The conditioner logs its own charge history to internal storage, tamper-evident but readable through the service port without breaking the seal. The left-hand unit reported four complete conditioning cycles since manufacture. The right-hand unit reported one. Neither number proved anything alone. Together with everything else they were beginning to compose a sentence.

Coolant residue. She pulled the inspection port and looked for the faintest trace of thermal paste displacement, condensation marks inside the housing, the ghost a heat sink leaves after it has cycled hot and cold. The right-hand unit was clean. The left-hand unit showed residue bloom on its cold plate: powered, loaded, cooled, repeated. Someone had run compute on this appliance before shipping it to her.

Then the appliance's own thermal wax: a small pellet in its custody logger, calibrated to melt at fifty degrees, a physical fuse against exactly the lie she was holding in her hands. The thermal strips she had cut from the crate lids three weeks earlier recorded a history in bands. This pellet answered a narrower question. If the unit exceeded its rated transport tem-

perature, the wax flowed into a catch channel and no firmware could reset it.

The left-hand unit's wax had flowed.

She sat back on the stool and looked at the two cases for a while. One clean, one warm. Vesper's supplier and a Lattice reseller, each side sending its own pre-run appliance to Monday night's ceremony, each assuming the other's shipment was honest. Three weeks ago a warm appliance had arrived here by accident or arrogance and certified a trial nine minutes early, and everyone had called it clock skew. Now the same signs lay on her bench in duplicate, sent openly, as if the practice were normal, as if the market had already absorbed the idea that witnesses arrive pre-warmed and only the details remained negotiable.

She wrote the findings into the custody log in her flat engineer's hand: filter dust, cycle count, residue bloom, wax state, times and initials. Then she signed it, because a finding you will not sign is a rumour.

Esteban found her there at noon. He read the log standing up, twice.

"You can reject custody," he said. "You know what that means."

"I know what missing the ceremony means." Lucia closed the log. "If Tuesday slips, Alero breaches the settlement schedule, the auction collapses, the tenancy revenue goes with it, and the campus loses the budget line that keeps the legitimate labs running through winter. I can count. That's most of my department."

"And yet you've written this down."

"Because it's true, and because I'd rather be unemployed than complicit in a receipt." She stood. "I'm not rejecting cus-

tody yet. I'm quarantining both units pending ceremony review, and I'm telling you now so you hear it from me and not from a lawyer."

He rubbed his eyes. For a moment he looked every year of his age and then some, an energy executive who had sold tenant confidentiality as infrastructure and was discovering what else came bundled. "There's something you should see," he said finally. "In procurement. Since you're the one writing things down."

The facilitation fee sat three folders deep in the facility's records, dated five weeks back, paid to a scheduling consultancy in Buenos Aires that owned, as far as Lucia could tell, a laptop and a post office box. The invoice line read *window alignment services*. The amount was obscene for consultancy and modest for what it bought: confirmation, buried in a routine amendment, that the root ceremony would open its window at the hour the largest auction needed it to.

Ordinary graft. No faction membership, no ideology, just money finding the loosest screw in the building. But it explained why the window sat so tight against settlement, why Esteban had been told *settlement drove the date*, and why nobody in the chain had thought the answer needed to be true.

"The board will want my head for showing you this," Esteban said.

"The board can want," Lucia said, photographing the invoice. "It's dated. It's signed. It exists. Everything else this week is somebody's story about the future. This is a purchase order for the past."

Back in the intake room she locked both appliances in the quarantine cage, logged the cage key into the safe, and stood a moment listening to the building hum around its secrets. Two

machines, warm and cold, waiting to be asked on Tuesday what they had witnessed. She knew their whole biography: filters, cycles, wax. Nobody had needed to hack anything. You just had to be the person willing to weigh the crate.

Chapter 22: Human Witness

They told her Martin was alive on Saturday afternoon, and let her see him that evening, in a converted storeroom at the back of the admin building with one window, a cot, a desk covered in printed chronologies and a guard outside who played a word game on his phone.

Vesper protection, the escort said, as if the phrase explained the lock on the outside of the door. Sofía supposed that from certain angles it did.

Martin looked thinner and bearded and completely awake, the way he looked during a long incident, except that incidents end. He saw her expression and raised a hand before she spoke.

"Before anything else. Yes, I know how it looks. Sit down. I have about twenty minutes before my keepers decide we've had enough reconciliation."

"You disappeared."

"I was taken." He said it flatly, without theatre. "After I tried to sell the mixed record twice, once to each side, Lattice went looking for pressure points. They found my daughter's school account. Recovery path, password reset, attendance

portal, the whole ordinary apparatus of a nine-year-old's week. No threats. Nothing anyone could take to police. Just her schoolwork suddenly visible to strangers, and a message that her safety was my responsibility now." He turned a page of his chronologies. "So when Vesper offered protection, I got into their car. Protection and captivity turned out to share a vehicle."

"And the mixed record." Sofia sat opposite him across the desk. "Tell me what it is, exactly."

"A continuity file. Everything I witnessed honestly for fourteen months as an Alero human witness, cross-referenced against everything I could prove about both frauds. The pre-minted receipts. The warm appliances. Vellum's derivative labels. Murmur's variant laundering. And the crossings, where both sides' fabrications met and certified each other." He pushed one of the printed chronologies towards her. "I built it because exposing either side alone would just hand the market to the other. Both at once, or neither. Mutual assured disclosure. It was elegant."

"Was."

"It failed," Martin said, "because I tried to sell it. Twice. That's the part of me you're allowed to despise. I found the worst thing happening inside the only market I've ever trusted, and my first instinct was to price it. I told myself neutrality needed funding. I believed myself for almost six days." He looked at his hands. "You were at my handover, the last good week. Ask me why the record is still intact and the answer is embarrassing: because nobody would meet my price, and because every day it sat unsold it became less a product and more a testimony. I'm not going to pretend the second motive arrived first."

Sofia read the chronology page while he watched. His work, unmistakably: the same care he brought to library maintenance, timestamps reconciled across four clock systems, every claim sourced, every uncertainty marked. In another life it was a changelog.

"I read the Nadia Ortiz thread," she said.

His face changed more than it had when he described the locked door.

"Then you know the part of this that doesn't fit into either side's deck. Her assistant was wrong about the cause. She was right about the failure. I rejected both because forty reports before hers had trained me to recognize confidence as waste." He turned one of the pages over, blank side up. "After we fixed it, people sent the thread back to me as proof I should accept everything. The next week we received ninety-three reports. One was real. I spent my daughter's Saturday finding it."

"And your other maintainer left."

"For a month. Then for good." He said it with the restraint of somebody naming a death outside the family. "She got an internal job maintaining an inventory system nobody can submit reports to. Better money, closed queue. Vouch lost the only person besides me who understood why half its ugly decisions were made."

"So you became an Alero witness."

"Yes. Because suddenly the market paid for the exact thing the commons expected for free: my ability to look at a polished claim and know what question would make it fall over. I told myself the boom had not displaced me. It had discovered my price." He looked around the locked storeroom. "That story was true for almost a year."

The guard outside made a word and the game chimed ap-

proval.

"What changed?" Sofía asked.

"The volume. At first I read every sealed report. Then I read the reproduced effect and sampled the path. Then I read the evaluator summary and checked the exceptions. By spring the summaries were written by one system about work done by another system and my job was to decide whether the disagreement score looked human enough to deserve my name." He rubbed his eyes. "I am very good at this, Sofía. Better than most people Alero could have hired. I still could not keep up. That's the humiliation nobody markets. Expertise doesn't make the day longer."

"Why keep signing?"

"Because every so often the queue produced Nadia. Something real, strange and important that no human had noticed. Refuse the system and I bury her again. Accept the system and I spend the commons one report at a time. There was no clean sceptical position left, only different ways to ration attention." He tapped the chronology between them. "So I built this. One record that would force both factories to slow down long enough for a person to read. Then I put a price on it, because apparently I learn the shape of a trap faster than I learn not to enter one."

"The crossings are the spine of it," he said. "Neither side can expose the other without self-incriminating. That symmetry is why everyone's still standing. But there's a hole in it, and the hole is why they let you in here, and why they'll both be polite to you until Monday night."

"The hole being?"

"They need your signature. Not mine. Mine is compromised; both sides have known since they started paying me.

Yours is clean. SRE of record on the original incident, operator of the Paraná node, name already on Alero's witness list since April through channels neither faction fully controls. If Sofía Alvarez validates the mixed record, it stops being Martin Vale's ransom note and becomes evidence." He leaned forward. "And yes. I am asking. My daughter's safety depends on both factions fearing this file's release. Your name makes the fear credible. That's the whole ask, and I hate every clause of it."

"What happens to the fear after Monday night?" Sofía asked. "If the ceremony runs and the mixed history joins the new root, does releasing your file even matter? The fraud becomes permanent infrastructure. Nobody's scared of a testimony about the past once the past is load-bearing."

Something moved behind his eyes that she recognized: the maintainer's shame of having missed the obvious issue until review. "The window closes either way," he said quietly. "Which means my leverage has a shelf life regardless of what you do. So I'll tell you the other thing, the one I should have led with." He took a breath. "Don't validate the whole record. Validate nothing you haven't personally checked. I built it honestly, but you can't know that, and your strength this week isn't believing people. It's intervals. Prove the impossible interval and let the record stand or fall on its own. If both factions watch you refuse to endorse it, the file gains more credibility than any signature could give it, because it will be the only thing left that nobody owns."

"That doesn't get your daughter clear."

"No. But it gets Tuesday honest, and honest Tuesdays are the only kind that ever made markets safe enough for people like her." He began squaring the chronologies back into their

folder, hands steady, workmanlike. "Twenty minutes is nearly up. Tell Lucia I said the wax test was well done. Tell Julián's mirror, if it survives, that its archive was right about the fork dates. And tell yourself, before someone in a bad suit asks you to sign something satisfying tomorrow night: you've already written down what you can prove. Read it out loud. Nothing more."

At the door she stopped, hand on the frame.

"The mixed record. Where is it?"

"Three copies," Martin said. "Held by people who don't trust each other or me. You taught the whole region that trick on Wednesday night, driving to Paraná. I just did it first, worse, and for money." He almost smiled. "Go on. You have a quorum to break."

Chapter 23: The New Market

The settlement floor occupied the east wing of the hall, behind glass, and Sofia walked it Monday afternoon on her legitimate delegate badge while the ceremony crews tested cables around her.

Alero's product was receipts, but receipts were only the certificate. The market lived here: rows of terminals where brokers for buyers she would never see reviewed packages, bid and hedged. Screens showed lots she was not cleared to read, redacted to class labels and price movements. She kept to the visitor rail and did what she had come to do: look for one name.

She found it at 16:40 on a pending-settlement board, in a package labelled with a class-two finding and a buyer code that resolved, through nothing more secret than the hall's public rate card, to a European integrator. The package carried an evaluation attestation. The evaluator of record was Micaela Ferreyra.

Micaela, who ran delivery routes in Balvanera and Almagro. Micaela, whose honest evaluation work Mateo had noticed and praised, and whose name now sat, printed and pend-

ing, under a sale she had never seen, in a building five hundred kilometres from the streets she actually knew. Someone had lifted the attestation from Lattice's contributor records, where Micaela's evaluation improvements were logged and dated, and stapled her competence to someone else's inventory. Her work as a human sticker on another entity's sale.

Sofía photographed the screen with her personal phone, timestamp visible, badge in frame, because evidence you might need later deserved boring metadata.

Then she left the floor and called Micaela.

The only place in the east wing where a delegate could speak without becoming part of the brokers' soundscape was a service stair between the settlement floor and the loading corridor. Sofía sat on the concrete step with her badge turned inwards and listened to the call ring through the noise of Balvanera. Motorbikes, a shutter, somebody calling out an address. The ordinary machinery of the business whose owner was being sold here as an evaluator.

"I have ninety seconds," Micaela said. "The app has put every rider west of Pueyrredón into the same zone and I have six people arguing with a map."

"Did you evaluate a security package numbered 4471-C?"

The noise on the line changed. Not silence. Micaela had covered the microphone and walked somewhere else.

"Where did you get that number?"

"It's on a settlement board at Piedra Sola. Your name is under it."

"My full name?"

"Yes. Evaluator of record."

Micaela said nothing for five seconds. Then: "Read me the date."

Sofía did.

"No. That week I tested route reconciliation. Mateo sent me a queue with broken orders and asked which ones the tool recovered. I marked pass, fail or couldn't tell. That was all. There was no security package."

"What did you agree to when you joined the queue?"

"A contributor page. Payment per completed evaluation if they used it. It said results might be aggregated." Her voice had acquired the flat speed Sofía remembered from the repair workshop, the voice of a woman keeping panic from becoming another task. "I read it. I thought aggregated meant they would add my pass and fail counts to everyone else's. I didn't think it meant my name could travel."

"Did you sign an attestation?"

"I clicked a box."

"What did the box say?"

"Something about reviewing the submitted material."

"Which submitted material?"

"The queue." Micaela stopped. Behind her a rider asked whether Congreso was east or central tonight. She told him east, wait for the refresh, then returned to Sofía in a lower voice. "There were forty-seven tasks. You opened one, the tool gave you a proposed answer, you tried it against the dispatch twin and marked the outcome. Most failed for stupid reasons. Wrong street ranges, old invoice rules. One worked when I didn't expect it to. After that I kept going."

"Why?"

"Because one worked."

The answer came with enough shame that Sofía did not ask twice.

Micaela continued anyway. "You don't understand what

those queues feel like. Thirty wrong answers and then one catches something I missed, and suddenly the next one might be the useful one. So you run five more. Then you're already late and the work is still there and maybe one more repairs the hour you lost. They paid only for accepted evaluations. I never knew which ones were accepted. Sometimes money appeared two weeks later. Sometimes nothing."

Variable reward, Sofía thought. Not incompetence. A slot machine built out of piecework and professional hope.

"Do you still have the original queue records?"

"The tablet has history. The workshop took an image when they repaired it."

"Don't update it. Don't clean it. Don't open the contributor page from that device again. Ask Julián for a read-only copy, and write down what you just told me while the dates are in front of you."

"Am I in trouble?"

Sofía looked through the stairwell's wired-glass window at the settlement floor. Package 4471-C had moved one row higher while they spoke.

"Your name is in trouble," she said. "I don't know yet whether the rest of you is."

Micaela breathed out once, hard. "That's not funny."

"It wasn't meant to be. I need you to do one thing I should have asked in the workshop. Tell me the boundary of your work. Not what Mateo said it might become. Not what the contributor page called it. What did you personally do?"

This time Micaela answered slowly.

"I supplied delivery failures from my own system after removing customer details. I ran proposed fixes against a copy. I marked whether the copy behaved better. I did not inspect

how the fixes were made. I did not test security. I did not see package 4471-C.”

“Write exactly that. Date it. Send one copy to yourself and one to Julián.”

“And one to you?”

“Not yet. Yours first.”

A refresh alarm sounded on Micaela’s side. The six riders had become her life again. Before she hung up she said, “Sofía, the one that worked really did work. It found duplicate jobs we were paying twice. That part is true.”

“Write that too.”

The call ended. Sofía returned to the rail and made herself look at the whole board, not just one name, because the whole board was the actual discovery.

Around Micaela’s line, filling every queue and waiting list, the excluded operators’ work was everywhere. Findings from researchers priced out of Vesper’s human-provenance premium. Results from small teams running open models on rented hardware, exactly the collective Mateo drove hardware towards. Genuine findings, several of them, if the class labels meant anything: real flaws found by people the market called noise, queued for review that would never come because no human reviewer existed whose time they could afford.

Noa appeared at the rail beside her without ceremony, hands in pockets, reading the same board.

“You’re looking at your friend’s name.”

“I’m looking at all of it,” Sofía said. “There’s real work in those queues. Some of it is good. You’d call it contaminated by origin.”

“And Mateo would call it valuable by result, and neither of you is wrong, and neither position survives contact with

that screen.” Noa nodded at the board. “Look what access has become. Not knowledge. Placement. A name in the right queue is worth more than a flaw in the right software. Everyone down there is buying and selling the right to be believed.” She said it without satisfaction; there was no satisfaction available to her in being right this way. “Vesper built its premium on refusing this floor. And here I am, standing on it, because my alternative is watching the whole idea of provenance die inside a root migration we can’t stop either.”

“Then help stop it.”

“With what? A purge?” Noa turned to face her fully. “You want me to say it out loud, so let me. If Vesper could reach into that board tonight and strike every model-originated claim, every assisted evaluation, every ambiguous authorship, we would. You know it. It wouldn’t save the market; it would just make our exclusion permanent and profitable.” A pause. “And you won’t hand us the knife either, will you?”

“No.”

“Because it can be spun.”

“Because it’s not mine to sign.” Sofía said it evenly, in English, formal. “Micaela’s attestation is false. I can prove the sale predates any review she performed, if anyone ever asks her what she reviewed. That’s provable. Whether every model-originated finding in that queue deserves exposure is not something I know, and I’m not certifying what I don’t know. Not for you, not for Mateo, not for the board.”

Noa studied her for a long moment, and whatever she concluded, she kept.

“They’ll both offer you something tomorrow night,” she said instead. “Before the window. Everyone gets offered something before a settlement; it’s how settlements stay soft.

Whatever shape it arrives in, check which of your accounts it lights up.” She glanced once more at the queues full of unowned work. “The ones nobody offers anything for are usually the truest. Look at them. They can’t even get an offer.”

She left towards the lounges, exact as ever. Sofía stayed at the rail a minute longer, looking at Micaela’s name sitting quietly under someone else’s price, and at the queues of findings nobody would buy because their authors lacked the one thing for sale: placement. Access as extraction, operating smoothly, one day before the ceremony meant to make all of it permanent.

She wrote two lines in her notebook, dated them, and closed it:

Package 4471-C bears evaluator attestation M. Ferreyra. Attestation cannot describe any act she performed. Provenance of the attestation itself is now the question.

The middle of the market is people. Both factions are pricing them.

The cable-testing crews finished at six, and the hall settled into its overnight quiet. Before the pending board dimmed, package 4471-C changed state.

ELIGIBLE FOR ROOT COMMITMENT.

At midnight tomorrow, unless somebody stopped the clock, Micaela’s stolen name would become money.

Chapter 24: Root

The memory came back complete at midnight, as she had known it would, because she had spent ten days feeding it pieces and the whole of it had been waiting for the last one.

She was in her borrowed room in the guest block, lights off, wind working at the window seal, and she let it arrive.

The Proofbox job. The last one. Late 2023, three years ago, a rented flat in Córdoba with two desks and one heater. Mateo on his chair by the window, offering her something no client had ever offered: the right to refuse. *We can walk away from this one*, he said. *The cleanup isn't finished and you know what signing an unclean effect means. I'll take the hit with the broker. We'll eat rice for a month.*

And beside the offer, glowing on his screen where he had left it carelessly, deliberately, the way he left every temptation he ever offered: the bonus terms. A Bitcoin payment uplift, expiring at dawn. Not enormous even then. Enormous relative to rice.

She had signed the effect despite incomplete cleanup because the bonus expired at dawn, and she had told herself the signature meant the test passed and nothing else, and Mateo had watched her do it and said nothing, and that nothing had been the true beginning of everything between them: he sup-

plied the exits and she chose to stay inside, every time, and both of them called it a partnership until it wasn't.

The memory settled into its place among the other pieces, and she lay in the dark and understood that it changed nothing about Tuesday and everything about her. The midnight signature on her dormant credential, the one Lattice reused, the one Vesper's dossiers waved like a flag: it had always been hers. She had lit it herself. Everything since was people finding the light.

The knock came at 00:40.

Not the door code knock of facility staff. A visitor's knock, hopeful, which in a locked building meant someone had talked their way past reception. She dressed, put the notebook in her pocket out of pure habit, and opened the door on Noa, alone, holding a folder like a peace envoy.

"Five minutes," Noa said. "Then Mateo gets his five, then Martin's record gets its five. They're scheduling your conscience down the corridor. I asked to be first so you'd know what the baseline looks like."

"Vesper's offer."

"Vesper's offer." Noa sat in the desk chair without being asked, laid the folder on the bed, and did not open it. "Full legal regularization of the Banca Rocca account. Real counsel, real filings, amnesty-path applications where they exist, quiet settlement where they don't. Your exposure ends. In exchange: your witness signature tomorrow attaches to human-provenance receipts only, and Vesper preserves its premium through the transition." She paused. "That's the honest version. Here's the version you should hear from me before you hear it from anyone else. If you sign that folder, you become a line in our settlement records. Human-witness Alvarez, re-

tained, regularized. The day Vesper needs someone credible to certify that its inventory is clean, there will be a document saying we own your judgement."

"You're telling me not to sign."

"I'm telling you what signing means. There's a difference, and it's the only difference I have left." For a moment the exactness slipped and something older looked out. "I came into this market believing custody could be kept. I still believe it. What I've stopped believing is that our company is the keeper. Sign nothing tonight, Sofía. That's my actual advice, free of charge, against every instruction I was given."

She left the folder on the bed, unopened, and went.

Mateo's knock came at 01:15. He didn't sit. He stood in the doorway with his jacket on, road dust still on his boots, and made his pitch from the threshold like a man who knew better than to bring cargo into the room.

"One-time keys," he said quietly. "Enough Bitcoin to close every exposure you have, tonight, before the ceremony. And in exchange, all you do is receive. The raw publication goes out under Lattice's disclosure framework at dawn: every crossing, both frauds, everything Martin found plus what we confirmed. Alero's market collapses under its own paperwork. No signature of yours anywhere near it. You just stop being blackmail-able, and the truth comes out regardless of anyone's premium."

"And the wallet that receives your payment," Sofía said, "becomes a wallet Lattice can trace forever. Wired into your ledger the moment the coins move." She watched him not deny it. "You'd make me clean by making me yours."

"I'd make you safe."

"You'd make me early," she said. "It's the same mistake

as everyone else's, Mateo. Certifying tomorrow before it happens. Dawn doesn't need your help arriving, and I don't need to pre-mint my innocence."

He stood there a moment longer, and she saw him actually hear it, the doctrine he had driven north inside a rented pickup arriving home to its owner. He nodded once, said, "You froze the wrong card, back then. You never freeze the wrong thing now," and was gone down the corridor.

Martin's offer arrived last, at 02:30, and it arrived as it had arrived in the storeroom: the mixed continuity record, needing only her validation to become credible, needing exactly the endorsement she had already decided she could not give. His messenger was a tired Vesper escort who clearly wished to be anywhere else.

Sofía sat on the bed afterward among three unaccepted futures and did the arithmetic Noa had named without naming it.

Vesper's regularization would wire her into Vesper's settlement records. Lattice's Bitcoin would key a wallet into Lattice's traceable graph. Martin's record needed her signature on his custody, which was its own account, opened in someone else's name and balanced with her trust.

Every clean path she owned ran dark and unlit: the Paraná images in Rosario and Palermo, the notebook sentence, Rina's preserved fraud record, Julián's spreadsheet, Micaela's name photographed against a badge. None of it paid her. None of it protected her. All of it stayed hers.

The clean path was the one that lit no other account.

She turned off the lamp, and outside the wind kept testing the window seal the way it had all week, patient, certain that everything loose eventually gives, and the building held,

and somewhere below in the hall the ceremony clocks counted down towards the hour when someone would ask her, finally, to say only what she knew or everything they wanted.

Chapter 25: Forty-Seven Minutes

The root ceremony opened at 23:07, eleven minutes late, because a lawyer for one of the settling buyers insisted on reading a reservation into the record before anything else happened. Sofía learned about the delay from the schedule feed and used the minutes to walk her route one last time: witness corridor, observation gallery, the narrow stair to Lucia's intake office, where she had permission, in writing, to be.

The contractual shape was simple and unforgiving. Pending settlement, bank leg and stablecoin escrow both, created a forty-seven-minute window at the start of the ceremony during which the new root had to be proposed, witnessed by quorum, and committed. If commitment failed inside the window, everything reverted: receipts unmigrated, auction unsettled, the whole week's inventory stranded in custody limbo while lawyers fed. Forty-seven minutes. Alero's own contracts had built the cage; everyone present was now living in it.

Sofía took her seat in the gallery at 23:04 with her notebook closed on her knees and counted the room. Buyers' brokers behind glass below. Faction representatives in the side seats:

Vesper's delegation crisp and unhappy, Lattice's delegation relaxed in a way that cost them credibility every time the schedule slipped. Esteban at the facility table looking like a man watching his own collateral. Daniel on the video link from Seattle, small and pixelated on the hall display, entitled under Traceframe's customer contract to observe and saying nothing yet.

And in the witness corridor, visible through the interior windows, eight grey cabinets and their custody teams, plus the two quarantined replacements in their cage, standing under seal exactly where Lucia had locked them.

The first pressure arrived at 23:11, before the root proposal had even been read.

Two custody teams approached the quarantine cage with paperwork: an amended ceremony manifest adding the replacement appliances to tonight's witness pool. The amendment cited the settlement schedule's force majeure clause. Behind them came three people in suits whose badges were new enough to gleam, and a facility security officer who kept looking at his shoes.

Lucia was already there. She had been there since 22:50, which told Sofía that her route-walk had been matched step for step by somebody.

"The custody review is open," Lucia said, not moving from in front of the cage. "Both units are under my quarantine, logged, sealed, photographed. They join the witness pool when custody review clears them or when the director overrules me in writing. Not by amendment."

"The settlement window," one of the suits began.

"Is forty-seven minutes long and none of my concern." Lucia held out her hand. "Give me the amendment. I'll file it

with my findings attached.”

The finding went into the record at 23:14, read aloud by the ceremony clerk because Lucia insisted: filter dust, battery cycles, coolant residue, flowed thermal wax, two appliances powered before declared delivery. The gallery went very quiet. Vesper’s delegation discovered urgent interest in its own papers. One of Lattice’s representatives asked, too loudly, whether wax calibration records were even admissible, and the clerk, to his eternal credit, said everything was admissible, this was a custody hearing wearing a ceremony’s clothes.

The root proposal was finally read at 23:19. Thirty-five minutes left.

That was when they moved on Martin.

It looked almost gentle from the gallery: two of the new-badged men crossing to where Martin sat in the observers’ row, speaking to him, taking his elbow, guiding him towards the side exit with the firm courtesy of people removing a disruptive guest. His testimony, unvalidated, unowned, was scheduled never to be heard; whoever controlled his presence controlled the last independent account in the room.

Sofia was moving before she decided to, down the stair, across the floor, and what stopped the removal was not heroics but custody procedure, arriving in the person of the corridor guard, who put himself between the suits and the door and recited, audibly and beautifully, the facility rule every person in that room had signed on entry: no principal, witness or observer leaves the hall during an open ceremony without a logged custody transfer countersigned by the facility duty engineer.

Duty engineer. Everyone looked at Lucia.

"I'm busy," Lucia called from the cage, filing papers. "Queue forms at the intake office. Transfers processed in order received."

Martin sat back down. The suits conferred. Twenty-nine minutes left, and nowhere in the hall was anything happening except reading, filing and the slow grinding of procedure against intention.

Sofía returned to the gallery and opened her notebook to the page written in ink days ago in Rosario. She had independent Traceframe timing for the critical intervals, federated clocks, the Paraná quarantine, Julián's fork dates. What she did not have, and would not pretend to have, was any basis to label individual claims in the migration set true or false. She could not say which zero-days worked. She could not say who deserved payment. Every faction in the room could give her words that would; none of the words existed yet.

At 23:31 the ceremony clerk called the objection period, and the president of the ceremony looked around the hall and asked, in the flat voice of a man inviting catastrophe, whether any participant wished to state a formal objection before commitment.

Forty-seven minutes had been designed to make this moment impossible to survive. Twenty-three remained. Sofía stood up.

Chapter 26: Exception

“I have an objection,” Sofía said, “and I want it recorded before any commitment vote.”

The clerk recorded objections in the order received, which meant she gave her name, her capacity, delegate of Traceframe, customer of record, and watched it enter the ceremony log at 23:33. Twenty-one minutes left.

“State the objection.”

“During a bounded interval, the required witness environments for the certifications under migration did not coexist in time. Therefore no valid quorum could have formed for those certifications. I am not objecting to every receipt in the set. I am objecting that continuity cannot be established across this interval, and asking that commitment be paused pending proof.”

She put the first page under the document camera. Not a conclusion. Four rows.

“This is the expected order for certification 7C,” she said. “Commitment, environment creation, test execution, witness observation, receipt. Five events. If anyone disputes that order, say so now.”

Nobody did. The room understood product specifications better than guilt.

She uncovered the next column. The receipt had committed at 04:13:08. The environment had registered ready at 04:22:41. Test execution began twelve seconds after that.

“Your control-plane record,” counsel for one buyer said. “Traceframe’s clock.”

“Yes. So I treated it as untrusted.”

The second page held the same events as recorded by the image registry, the storage write path, the billing interval and the São Paulo audit sink. Four systems, four clock owners. Their stamps disagreed by milliseconds. None moved the environment to before the receipt.

The buyer’s counsel leaned towards his microphone. “A replicated bad source is still one bad source.”

“Correct. Which is why I checked the node’s monotonic counter.”

Third page. The number did not give wall time and she did not pretend it did. It gave order: the worker had been alive for a fixed count when the witness pods signed and a larger count when the environment first consumed power. A wall clock could be wrong by nine minutes. A monotonic counter could not lose five hundred and seventy-three seconds without the node restarting, and the hardware record showed no restart.

“Boring explanation one was clock skew,” Sofia said. “It failed. Boring explanation two was a reused environment name. The immutable creation record and image digest exclude that. Boring explanation three was a stale receipt. The receipt commitment is unique and appears in Alero’s migration set tonight. If it is stale, continuity already fails for a different reason.”

The president looked down the tables. “Does any party con-

test the source records rather than their interpretation?"

Vesper's counsel asked to inspect the hashes. Lattice asked whether the Paraná image had remained in Sofía's sole custody, putting a little emphasis on *sole*.

"No," Sofía said. "That would make it weaker."

She put the quarantine sheet up. The node had gone read-only at 02:06 on Wednesday. Its images were created under the facility camera, hashed once in the aisle and once in the car park, then separated. One copy entered a Rosario safe deposit under Tomas Alvarez's receipt. The other entered a locked drawer at Traceframe's Palermo office under the building guard's log. The hashes on the two copies matched the hash the ceremony clerk had received before the objection period.

The clerk checked. It took forty-two seconds. In a room running out of time, everyone listened to keys and paper.

"Match," he said.

Sixteen minutes remained.

The fourth page was Julián's fork chronology. Sofía did not call it proof of the nine-minute case. It was a different record held by different people for a different purpose: the community mirror's commit times for the claim now sold as package 4471-C. The discussion predated Alero's first certification by nineteen days. The claim then crossed from a branch accepting disclosed assistance to a branch banning model-origin reports, losing its authorship trail while keeping the same inherited error.

"That proves a forum discussed similar work," a Vesper representative said. "It does not prove identity."

"Agreed," Sofía said.

The representative stopped, deprived of the argument he

had prepared.

“The inherited error supports identity but does not establish it alone. I use this page for one narrower fact. Alero’s receipt cannot be evidence of when the claim first existed. The independent archive shows the market discussion was already old when the witness system dated its birth.”

She placed Micaela’s name beside the package number and nothing more. She had not received Micaela’s written boundary yet and would not quote a phone call as if it were an attestation. The board photograph established only that an evaluation bearing her name was pending settlement tonight. The timestamped queue records, if preserved, could establish the rest later.

“Then why include it?” the president asked.

“Because commitment tonight would convert an unresolved attribution into paid history. Pausing preserves the question. Committing answers it without evidence.”

The fifth page was Lucia’s, already part of the ceremony record: declared delivery times, measured weights, filter dust, battery cycles and flowed wax. Two ordinary appliances had worked before their paperwork said they arrived.

Lattice’s representative objected that the wax showed heat, not computation.

“Sustained heat above fifty degrees,” Lucia said from the floor. “The cycle log shows powered conditioning. The filter shows airflow. The cold-plate residue shows load and cooling. Each fact alone has several causes. Together they exclude sealed transport at declared condition. That is all my finding says.”

Sofía waited until the clerk had entered Lucia’s exact wording. Then she laid the transparency over the five pages.

Three crossings appeared. She had marked them by interval, not faction. At the first, a Vesper-supplied witness signed for an environment before Lattice's scheduled run. At the second, a Lattice witness signed inside the warm replacement window. At the third, both populations contributed signatures to a quorum before the target existed. Each receipt was individually valid under the migration rules. Taken together, they described witnesses that had never shared the event they certified.

"You are alleging collusion," somebody behind the buyers' glass said.

"No. Collusion would be one explanation. Independent pre-minting is another. I cannot distinguish them from these records and I do not need to. Quorum requires coexistence under either explanation. These witnesses did not coexist."

She removed the transparency, and the five pages became ordinary records again: clocks, a mirror, a settlement board, an intake log. No single page contained the plot. That was why the join mattered.

"I can't tell you what the exploits are worth," she finished. "I can't tell you who authored them or whether they work. Those questions stay exactly as open after tonight as before. What I can prove is smaller and harder: during these intervals, quorum was impossible. A market built on quorum cannot commit continuity through intervals where none existed."

The room began to talk at once, and the president spent two minutes restoring silence.

Then the counterattack, timed like everything else that week.

A Lattice representative rose with a tablet and an apologetic expression. "Before the panel weighs the objection, we should

establish the objector's standing. The witness credential held by Ms Alvarez, issued in 2023 and enrolled on Alero's witness list this April, has been active on four sealed runs in Neuquén province." He let the room absorb it. "We don't allege anything. We simply note that a person whose signature appears throughout the contested history may not be best placed to object to it."

The gallery turned towards her, and Sofía felt the whole week pivot on the moment, everyone waiting to see if the woman from the incident would defend herself.

"The objection doesn't depend on me," she said. "Strike my name from the record and the evidence stands unchanged. Federation timestamps don't care who reads them. Fork commits don't care who found them. Thermal wax doesn't care who melted it. I signed nothing in Neuquén except one job three years ago whose cleanup I've already disclosed to my family and will disclose to any tribunal that asks. But the interval is true whether you believe me or not, and that's why it survives your point. You haven't caught the objector. You've demonstrated that the only weapon left is biography, because the timestamps are clean."

Silence, then the small rustle of people rereading their own notes. The attempt to burn her identity had failed precisely because she had never built her claim on being anyone.

Lucia spoke next, from the floor, formal as a maintenance report: custody rejection of both replacement appliances, grounds as filed, no further evidence offered. Martin Vale followed, standing unassisted: "I created the mixed continuity record. I tried to sell it twice. Every entry in it is mine." Three limited human statements, each signed to only what its signer knew, now aligned on the record.

At 23:38, the hall display switched itself to the Seattle video feed without anyone's permission, because Daniel Cho had asked the clerk for the floor.

"Traceframe waives confidentiality," he said. "Effective now, on the record." He read steadily, a man choosing each sentence like a load-bearing beam. "The customer exception under which the contested tests ran was executed with documentation our own review has failed to validate. The exception files corroborate the objector's interval. I authorized the exception; I'm accountable for it; and I'd rather answer for a bad contract than let the contract decide what happened here." The feed cut back to the hall. Somewhere below, a broker said, very quietly, "there goes his career," and someone else said, "there goes the last excuse."

Nine minutes left. An assistant to the ceremony panel proposed a resolution: attribution of the entire corruption to a state-level programme connecting Vesper to official bilateral activity, confidence high, documentation attached. It explained everything. It would freeze Vesper's network in forty-eight hours. It was also circular, citing Observatorio Sur's briefing, which cited investor travel, which cited the programme, which cited the briefing. The assistant presented it as a service.

Sofía read it once, felt how satisfying it was, and rejected it aloud for the record: "The inference exceeds the custody of its evidence. Decline." She saw Noa, across the gallery, close her eyes briefly, a woman handed absolution and refusing to take it secondhand.

At 23:49, Esteban Roldan took the facility microphone.

"The institution pauses continuity," he said. "Neither acceptance nor destruction. All history enters escrow pending

external audit, technical and financial. Settlement does not proceed within the window.”

The window expired at 23:54, unmet. The ceremony log closed its forty-seven minutes with the root uncommitted, and the largest auction in the market’s history became, in the space of one clerk’s keystroke, unsettled.

Nobody cheered. The factions’ delegations sat motionless, twin ruins wearing each other’s signatures. Sofía closed her notebook and discovered her hands were shaking, not from fear, from the specific exhaustion of having carried a narrow thing through a wide room and refused, eleven separate times, to make it bigger.

On the video link, Daniel’s pixelated face did something that might have been relief. On the floor, Lucia began logging the cage seals into escrow, methodical, unhurried, first person in the chain of custody of whatever happened next.

Chapter 27: Custody

The window had been dead for six minutes when the badges came for the cage.

Lucia saw the group form before it moved: four of the new-badged suits, two custody contractors, and a lawyer she recognized from the Vesper delegation walking slightly behind, the way people walked behind plausible deniability. Their target was not subtle. The rejected appliances sat in quarantine under her seal, and whichever faction collected them first owned the only physical proof of its rival's fraud. Evidence, like everything else in that hall, was inventory if you could move it.

"Quarantine is not storage," Lucia said, placing herself between the cage and the lead contractor with her clipboard raised like traffic control. "It's an evidentiary state. Nothing leaves until escrow transfer is executed, and escrow transfer requires my countersignature, the duty engineer's and a facility escort of record." She checked her watch with theatrical calm. "It's midnight. The duty engineer is me. I decline."

"The settlement has failed," the lawyer said. "The contract's survival clause reverts custody of tenant property to the tenants."

"The ceremony hasn't failed. It's paused into escrow by the director's own ruling, which means facility custody continues,

which means that cage is mine until audit says otherwise." She turned to the contractors, who were employees of firms she knew and could starve of work. "Gentlemen. Your companies hold facilities contracts with this campus. Read your own indemnity clauses about breaching evidentiary quarantine, then decide whether tonight's hourly rate is worth the next five years of litigation."

The contractors looked at each other and became very interested in their shoes.

The second move came at 00:20, and it came for Martin.

Different men this time, Lattice-side, quieter, better prepared: they arrived at the observers' row with a document asserting protective transfer of a person at risk, medical grounds, his own safety, all signatures in place but one. Sofia, still in the gallery, watched the same flaw repeat itself. They had paperwork. What they lacked was a receiving custodian, because no one wanted to sign a custody gap into existence while auditors were being booked.

Martin solved it himself before anyone could force it. He stood, faced the men, and said, clearly enough for the nearest microphone to carry:

"I'm staying. And before anyone drafts anything else, note the time: the mixed record left my exclusive control an hour ago. Copies are now held by Alero's external counsel, Traceframe's delegate, the ceremony clerk's escrow and one other party none of you have identified yet, because I distributed them before I flew south. Release it now and it harms both of you equally and protects neither. My daughter's school account will be closed and moved tomorrow; the people doing it don't answer to either faction. There is nothing left to buy from me and nowhere left to sell me. Go home."

He sat down again, a maintainer returning to his chair after closing a ticket, and the men withdrew to wherever quiet failure went.

By half past midnight it was over without a single hand laid on anything. Sofia stood at the gallery rail with Noa on one side and the emptying hall below, watching Lucia log each refused transfer with the satisfaction other people took from winning races.

"They'll be back as lawyers instead of suits," Noa said. "Days of this. Weeks. Escrow fights are worse than ceremonies."

"I know."

"You could still have had the regularization." Noa said it without pressure, an observation offered to the room. "You'll face the tax question with no one's help now."

"Yes."

"And Daniel Cho ends his career on your interval."

"He ended it on his contract. I just supplied the timestamp." Sofia watched the brokers' glass go dark section by section. "What happens to you?"

"Disclosure I signed tonight. Vellum's outputs become evidence against my own market by morning." A tired almost-smile. "I called it a weapon, not an author. Turns out evidence is a third category nobody briefed me on." She gathered her coat. "For whatever it's worth, at Aeroparque I told you that knowing wasn't the same as proving. You were already doing the proving. I just hadn't caught up."

She left. Below, Esteban signed the escrow schedule into existence, and the ceremony clerk stamped the last page, and

Martin remained exactly where he was, guarded most effectively of all by the fact that removing him would create a custody gap bigger than the one Sofía had spent forty-seven minutes proving.

Settlements frozen. History escrowed. The market awake at one in the morning in a hall in northern Patagonia, discovering that certainty was the thing it had actually been selling all along, and that it did not know how to price its absence.

Lucia locked the cage, pocketed both keys separately, and turned off the witness corridor lights on her way out, eight cabinets and two quarantined appliances settling into the dark together, holding their clocks.

Part Four: Unsettled

Buenos Aires. The day after settlement did not happen.

Chapter 28: No Price

Noa had rehearsed the handover on the flight north and it went no better for the rehearsal.

The disclosure room in Vesper's Madrid counsel's office was small, beige and soundproofed, the kind of room companies built when they expected to say things that would be quoted back to them. Across the table, two of her own firm's lawyers, an external auditor and a recording notice she had helped draft standards for, years ago, in a better market.

Vellum's outputs. All of them: the fingerprinting corpus, the hostile cases generated against rival systems, the dead ends fed into Lattice's queue, the derivative labels applied to genuine discoveries, including one valid Murmur finding suppressed to protect the price of a human-origin package she had personally approved. Everything the instrument had touched became evidence against the market the instrument had been built to defend, and she was the one who signed the chain of custody, because she was the only person alive who understood all of it, which was its own verdict.

"The premium is gone," her managing partner said afterwards, not unkindly. "Human provenance as a price basis is over. The auditors will want to know if any of our inventory survives review, and I can't tell you yet whether the company

does.”

Noa looked out at Madrid through glass that needed cleaning. “The inventory that survives,” she said, “will be the parts where a named human can explain every step. That was always the actual asset. We just kept pricing the label instead.” She signed the last page. “I’d like my notes preserved with full timestamps. If this field has a future, someone will need to know exactly when we knew.”

The auction never settled at all.

By Tuesday night the bank compliance legs were frozen across three jurisdictions, each institution citing the others’ uncertainty, a perfect circle of nobody moving first. The stablecoin escrow stayed visible on its public contract, disputed funds locked in plain sight, buyers demanding refunds through channels that required the escrow’s own arbitrators to agree on what had been sold, which they could not, because what had been sold, certifiedly, provably, had never coexisted in time.

Vesper lost the premium within hours; origin labels stopped commanding any spread the moment the market noticed both sides’ labels were forged. Lattice lost payment on volume inventory worth, by its own internal estimates, more than the company’s runway, findings real and fabricated alike stranded in escrow because nobody could establish which was which without the audit neither side controlled. Alero, facing contractual breach from every direction, did the only thing left: invited external technical and financial auditors into Piedra Sola with powers Esteban Roldan signed at four in the morning in a shirt he had worn for two days.

No raid came. That was the part Sofía found most striking, reading the wire reports in Palermo on Wednesday. No government validated the conspiracy, no police carried out appliances, no headline used the word cartel. Ordinary contracts became unbearable instead, quietly, the way load-bearing things fail: settlement froze, premiums evaporated, escrow held, and the entire edifice discovered it had been standing on agreement rather than steel.

Micaela Ferreyra's frozen settlement package surfaced in the auditors' first week, exactly as Sofía's photograph said it would, and the questions arrived in Balvanera by courier, formal letters asking Micaela to explain an evaluation attestation bearing her name on a sale she had never reviewed.

She came to Palermo with the letters in a shopping bag and sat at Sofía's kitchen table looking smaller than her own delivery floor.

"They think I'm either a fraud or an idiot," she said. "My riders heard *investigation* and half of them won't take shifts. And the worst part, the part I can't get anyone to understand, is that I don't know either. Someone used my work and I can't even prove what my work was."

"You can," Sofía said. "Not everything. Just exactly enough."

From the shopping bag Micaela took the page she had written after their call from Piedra Sola. It was creased into quarters and stained at one corner with chain grease. Four sentences, the boundary Sofía had asked for: delivery failures supplied, proposed fixes run against a copy, outcomes marked, no security review performed. Micaela had dated it and sent copies to herself and Julián before midnight.

"This is what you remembered," Sofía said. "Now we find

what the machines remember.”

Julián’s read-only image sat on an external drive between them. They opened the dispatch history first. Forty-seven evaluation tasks, as Micaela had said. For each one, the tablet retained the time she opened it, the synthetic route set it touched and the outcome she chose. Thirty-two failures. Nine inconclusive. Six passes. Package 4471-C appeared nowhere.

“That proves I didn’t see it?” Micaela asked.

“It proves this device has no record of you seeing it. We keep the claim at that size. What other device could you have used?”

“None. The contributor page only worked from the tablet after Mateo enrolled it. I complained because I wanted to do queues on my phone between calls.”

“Can you prove it only worked here?”

Micaela searched her messages and found the complaint, sent to contributor support, and the answer: evaluation access bound to the enrolled tablet for custody. A restriction sold as trust had preserved her defence.

They added it to the timeline.

Next came the proposed fixes. Sofía expected them to contain the same inherited error as package 4471-C. They did not. She checked twice, then made Micaela watch her remove the theory from the draft.

“But that would connect them,” Micaela said.

“It doesn’t.”

“It almost does. The component name is the same.”

“Almost is how your name got onto their package.”

Micaela pushed her chair back and walked to the window. Below, a courier argued with a building porter about which apartment had ordered lunch. The argument rose four floors,

complete and useless.

“Do you know what I kept thinking?” she said. “When the first letter came, I thought if I ran the queue again I might recognize something. One more prompt, one more test. I thought the tool could tell me whether I had done the work.”

“Could it?”

“It gave me three different explanations. The third sounded best.” She folded her arms. “I nearly sent it to the auditors.”

“Why didn’t you?”

“Because it said I had verified a sanitization boundary, and I didn’t know what that was. I was about to look it up so I could understand the sentence I was supposedly remembering.” She came back to the table. “That’s the idiot part. Put it in.”

“It isn’t required to prove the attestation isn’t yours.”

“Put it in anyway. It’s required to prove which part is.”

So they wrote two sections. What the records supported, and what Micaela admitted without being able to support it further. The first listed the forty-seven tasks, the bound tablet, the six accepted outcomes and the absence of package 4471-C. The second said she had shipped a generated component she could not fully explain, had relied on workshop repairs without reviewing each change and had nearly used another generated answer to reconstruct work she did not remember doing.

Micaela read the second section three times.

“They’ll use this against me.”

“They might.”

“You could say something nicer.”

“I could say something you can’t sign.”

Micaela took the pen.

“And that’s enough?”

"It's the only thing anyone will believe, which makes it more than enough." Sofía turned the sheet around. "Nobody in this whole mess has produced one honest signature. The market asked everyone to certify more than they knew. You're going to be the person who didn't."

Two hours after she arrived, she wrote the final boundary in the same practical hand that planned delivery routes: *this part I checked, this part I could not check and shipped anyway, this attestation is not mine.*

When she left, Sofía read the account once more and recognized it for what it was: the first honest signature the market had ever produced, written at a kitchen table by a woman the factions had considered raw material.

She scanned it, sent a copy to the auditors through counsel, and filed the original in the folder that now held every narrow thing she owned: Rina's preservation record, Julián's spreadsheet, Lucia's intake findings, her own notebook sentence.

The folder had no lock. It didn't need one. Its contents were worthless to everyone except the people who wanted the truth to have dates.

Chapter 29: A Named Person

The statement was four sentences long and took three weeks of argument to write.

Sofía signed it in counsel's office on a Thursday, initialling each sentence separately because her lawyer insisted and she agreed, the way she had come to agree with everything narrow. The exception statement: that during the stated intervals, witness environments did not coexist, and quorum could not have formed. Nothing about the exploits' value. Nothing about authorship. Her own financial conflict disclosed at the end, in plain words, because counsel said undisclosed conflicts destroyed narrow statements and she was done financing other people's versions of herself.

What she refused, she refused on the record. The panel's final report wanted attribution connecting Vesper to official bilateral activity; she declined in writing: *the inference exceeds the custody of its evidence*. It wanted validation that Lattice's underlying findings worked; she declined: *I never tested them and will not certify what I cannot reproduce*. Both refusals went into the file beside her signature, which was the point. A named person may sign only what they can explain.

Noa left the process with what she called partial defeat and Sofía privately thought was the honest outcome: evidence against Lattice preserved, delivered, usable; Vesper's own disclosure signed first, its premium dead, her career a question mark measured in disclosures. Mateo left with his mirror image: evidence against Vesper intact, Murmur's genuine findings acknowledged in the audit, and the reused identity he had measured himself against burned, unusable, refused by every registry that touched it. Neither faction got its clean victory. Both got exactly as much truth as their conduct deserved, which is the most a process can deliver.

Rina Belluzzi survived her internal inquiry narrowly and by procedure, which was to say completely: the preserved fraud record, filed before close of shift on the day everything began, proved she had followed process while exercising judgement inside it. Her preserved record became part of counsel's file in Buenos Aires without her name needing to appear in any newspaper. She sent no message about it. She had been a clock; clocks don't comment.

Daniel Cho remained accountable for the exception at Traceframe's independent review, by his own insistence, in a statement whose first line read *I approved it* rather than anything longer. The review suspended Sofía for two weeks pending the same inquiry, full pay, her incident file finally readable by the people reading it. She spent the suspension doing nothing in particular and found, to her surprise, that she was bad at it.

On the thirteenth day they called her into the review. Daniel sat at the far end of the video table with independent counsel beside him and no company background behind his head, only a white wall. He had refused the option to submit answers in writing.

The panel's first proposal was a prohibition: no model-originated security work inside Traceframe's verification services until a human reviewer had inspected every changed line.

"That would have prevented this incident," the chair said.

"No," Sofía said. "It would have moved the lie. People would sign that they inspected the lines. The queue would keep growing. Review theatre would become policy."

"Then what do you recommend?"

She had brought Micaela's account. Not the package bearing her name, not the auditors' interpretation. The page in Micaela's own hand.

"Separate the questions. Who defined the objective? Who can reproduce the effect? Who understands the acceptance criteria? Who can stop deployment? Who will maintain it after the person who prompted it leaves? Those may be different people. Name them. Don't compress them into one author field."

An engineering director objected that nobody could explain every line in a modern dependency tree. Sofía agreed.

"Explanation is not recitation. Tomas doesn't know every component in an anaesthesia station. He knows what it controls, what evidence says it is inside its safe range and when he must take over. Mateo knows every evaluator score in Murrur and could not tell me what happened to sealed material after a run. The difference isn't human versus model. It is whether ignorance has a boundary and somebody has authority to act at that boundary."

The youngest person on the panel, a staff engineer who had joined through an AI-assisted apprenticeship programme, asked the question nobody senior had.

“Doesn’t this rebuild the gate? If I need an expert’s name before I can submit work, people like me never get to become experts.”

“Nadia Ortiz found something Martin missed,” Sofia said. “She didn’t understand Vouch when she filed it. She understood the failure she reproduced and said where her understanding ended. That is enough to enter review. It isn’t enough to make volunteers review an unlimited queue for free. Access and attention are two different obligations.”

The labour representative asked who paid for the attention.

Nobody answered at first, which was the most honest moment in the hearing.

Then Daniel spoke. “The customer does. Or we decline the work. In the exception I approved, confidentiality hid the review burden and the contract priced only runtime. We treated human understanding as overhead. That decision is mine.”

The final policy took another week. It did not ban assistance, chat-generated code, repository agents or spec-driven implementation. It required disclosure of where each had acted, test evidence independent of the producing loop, a named stopping authority and funded maintenance for any system whose output entered a customer claim. Volume above the funded review limit waited. A green evaluator could not spend a human future it had not purchased.

Sofia disliked three clauses and signed none of them. The panel adopted the policy anyway. That too was part of named responsibility: her evidence could change a decision without making every later decision hers.

The tax question remained hers. Rina could not solve it; banks solve custody, not conscience. Tomas helped her find counsel without forgiving the lie immediately, and their

Sunday calls resumed at half their old warmth, which she understood to be the market price now, honestly struck, payable over years. Their mother remained outside the details entirely, which both siblings maintained like a shared production schedule.

Micaela's written account travelled further than anyone expected. Counsel attached it to filings. An auditor quoted it. When Traceframe's review asked Sofía to state her position on attribution standards, she submitted two pages that were mostly Micaela's structure wearing her vocabulary: a person may sign only what they can explain; assistance must be disclosed where it acted; uncertainty belongs in the signature block, not hidden behind it. The review adopted the language almost verbatim, which she suspected meant some reviewer elsewhere had reached the same shape independently, from their own kitchen table.

On the last afternoon of the suspension, Mateo came to Palermo unannounced and they drank mate at her table like the end of something rather than the resumption of anything.

"The collective got real," he said. "Fourteen people now. Actual witnesses, actual runs, disclosed origin, no premium. We're poor but we exist." He turned the gourd. "You'll never use us."

"No."

"Because I'm compromised."

"Because you're early," Sofía said, and he laughed, really laughed, at hearing his own word come back across the table with its meaning inverted and kept.

He left at dusk. She washed the mate cups, checked the folder once, all narrow things present, and stood at the window over the street where the delivery bikes ran their evening

circuits, each rider carrying an order to a person who had paid for a future that would, for once, actually arrive.

Epilogue: The Number After

The report arrived on a Tuesday, which Sofía decided was either tradition or irony and did not investigate further.

It concerned a fictional open-source project she followed casually, a small scheduling library with three maintainers and a quiet issue tracker. The report was AI-assisted and said so, in its own header: tool disclosure, model family, prompt strategy summarized. Reproducible evidence attached, a test case that failed before the fix and passed after it, runnable by anyone with an afternoon. And at the bottom, a human sign-off: a name, a date, and the sentence *I have reproduced this result and can explain the change*.

The project's longest-standing maintainer wanted to reject it on origin alone. His argument filled most of a thread: generated reports had consumed his volunteer hours for two years, he owed nobody the labour of reviewing machine output, and closing the gate entirely was the only sustainable policy he could afford.

Sofía read the thread twice, then the evidence once, then the disclosure header again, and entered the discussion the way she now entered every discussion: through what each party

could actually sign.

She asked the named person one question, publicly, where everyone could see it: explain the decisive step in your own words, at whatever length you need.

The answer came back the next morning: four paragraphs, slightly awkward, genuinely understood, describing why the scheduling boundary misbehaved and what the fix traded away to work. Not eloquent. Owned. The maintainer who had wanted to reject the report replied with a single word, *merged*, and a comment that made her smile at her kitchen table: *reviewing this cost me less than arguing about it*.

Elsewhere in the same week, smaller news that never made wire services. Micaela's dispatch tool ran its first full month under a disclosed-assistance arrangement maintained through Julián's repair community: documented components, named maintainers and one module listed openly as *behaviour not fully understood; change requires two-person review*, because naming the gap had turned out to be the whole trick. The unexplainable component finally had a maintainer instead of a hope, and Micaela had stopped hiding how much she did not know from anyone, which her riders reported, via the building's grapevine, had made her easier to work for, not harder.

None of that put the previous world back. Vouch's second maintainer stayed in her closed internal job. The backlog fell only after the project capped new reports at the number of funded review hours, which meant useful submissions waited beside useless ones and some newcomers went elsewhere. At Piedra Sola, the machinist who now drove the shuttle did not regain the drawing work that automation had taken. Better rules could stop a system spending tomorrow's attention.

They could not refund yesterday's years.

Ariel, the rider who had waited unpaid in the Almagro workshop, began taking one afternoon a week to learn the dispatch tool's test harness. He was not becoming a programmer in the heroic language of training brochures. He was becoming the second person who knew how to stop a bad route update before fifteen riders discovered it in traffic. Micaela paid for the afternoon. The queue ran slightly worse while he learned and became less fragile afterwards.

Julián himself sent her the mirror's quarterly summary without comment: archive intact, both fork branches still running, attendance down, uptime boring, everything a commons should be. She sent back one line, *received*, and they left it there, both of them adults who knew exactly what had been repaired and what had merely survived.

On Friday, the replacement Banca Rocca card arrived downstairs.

The buzzer went at eleven, the courier held out the padded envelope, and Sofía signed for it with her real name while the card's digital successor sat already active on her phone, provisioned the day the request cleared, waiting in her wallet app like a fact that preceded its own evidence. She stood in the doorway holding the plastic that had been rendered ceremonial by its own infrastructure.

Then she disabled the digital card, all of it, every token and scheduled payment, recorded the event with a screenshot and a timestamped note, and filed the statement in the folder with everything else narrow and dated.

She would use a bank again. She would even use the phone; the audit trail on that decision belonged to future Sofía, and current Sofía had learned better than to certify on her behalf.

But the act of disabling it herself, recording it, keeping the statement: that was the part no replacement product offered, and it cost nothing except the willingness to look like a woman who did not trust her own wallet, which by now was simply accurate.

The envelope went into recycling. The phone went in her pocket. Upstairs, the scheduling library accepted contributions again under rules everyone could explain, and somewhere north of the city, on shelves she would never visit, escrowed history waited for auditors patient enough to give dates their proper weight.

Sofía made coffee and opened her laptop to the merged thread one more time, not to re-read the fix but to look at the bottom of the report, where the whole argument ended and none of it mattered anymore:

a name beside the claim.